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A NEW RESULT ON THE RELATION BETWEEN DIFFERENTIAL-ALGEBRAIC REALIZABILITY AND STATE SPACE REALIZATIONS

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ABSTRACT

For analytic input/output operators, we prove that realizability by a (possibly singular) polynomial state space system is equivalent to the existence of an algebraic differential equation satisfied by the input output behavior.

1 Introduction

By an *analytic i/o mapping* we mean an operator

$$y(\cdot) = F[u(\cdot)]$$

described by a convergent generating series (or equivalently, by a Volterra series with analytic kernels). This is a very general kind of i/o causal operator, and it includes a large variety of nonlinear systems; the book [6] provides an introduction to such operators. To each such i/o map one associates its *behavior*, that is, the set of i/o pairs

$$w(\cdot) = (u(\cdot), y(\cdot))\,.$$

This is the observed data in an experimental situation. For *linear* systems one knows that realizability of a behavior by a linear finite dimensional system is equivalent to the existence of an "autoregressive moving average" representation

$$\begin{aligned} y^{(k)}(t) &= a_1 y(t) + \ldots + a_k y^{(k-1)}(t) \\ &+ b_1 u(t) + \ldots + b_k u^{(k-1)}(t) \end{aligned}$$

relating these pairs, or equivalently, a linear equation of the type

$$E\left(w(t), w'(t), w''(t), \ldots, w^{(r)}(t)\right) = 0$$

(as preferred in some of the recent system-theoretic literature, in particular the work of Willems and his school). In frequency-domain terms, the existence of such an equation means that the transfer function associated to F is rational.

It has been long been expected that some such relation between the existence of i/o equations (with E not necessarily linear) and realizability should also hold for nonlinear systems. Indeed, it was one of the main objectives of the work [10], [11] to show that there is indeed such a relation: in that case, realizability by what are there called finite-dimensional "k-systems" (basically a class of rational systems) was shown to be equivalent to the existence of an i/o equation with polynomial E. This work was applied in the context of identification problems, and stochastic versions where also studied (see for instance [7] and [3]). More recently, these results where extended to continuous-time bilinear systems and a theorem showed that realizability by such systems is equivalent to the existence of some E of a special form, namely affine on y (see [12]). Some authors, –see e.g. [4],– have in fact suggested that "realizability" should be *defined* as "existence of an i/o equation", as the natural definition from the point of view of differential algebra.

The main new result is as follows. We shall say that F is *realized* (by a singular polynomial state-space system) if there exists an integer n, polynomial vector fields

$$f, g_1, \ldots, g_m$$

on $\mathbb{R}^n$, and two polynomial functions

$$q, h : \mathbb{R}^n \to \mathbb{R}$$

such that the following properties hold: (1) for each i/o pair, there is some solution $x(\cdot)$ of

$$q(x(t))\dot{x}(t) = f(x(t)) + \sum_{i=1}^{m} u_i(t) g_i(x(t))$$

such that $y(t) = h(x(t))$ for all t, and (2) there holds the following *regularity* condition: For a C^ω pair generic in the Whitney topology, and for almost all t, the above solution satisfies that $q(x(t)) \neq 0$. Then we have the following result:

Theorem 1 *The operator F is realizable if and only if it satisfies an algebraic i/o equation (E polynomial).* ∎

In addition to single i/o maps, it is also natural to study *families of i/o maps*, defined by a *family of convergent generating series*. To study a single i/o map may seem to be natural as a formal description of a *black box*, but in general, a system may induce more than one i/o map. For example, a system described by an ordinary differential equation on a manifold may induce infinitely many i/o maps, each of them corresponding to some initial state.

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One should study all the i/o maps induced by the system simultaneously than to study them individually, unless a fixed initial state is of particular interest. This leads to the concept of a family of i/o maps. One question arises naturally: when can a family of i/o maps be realized by *one* state space system? i.e., when can all the members of the family be realized by some singular polynomial system, each of them is associated to some initial state of the system? The result we have so far obtained is that a family of i/o maps is realizable if and only if all the members of the i/o maps satisfy an i/o equation simultaneously.

There has been much work attempting to characterize when the existence of an algebraic equation implies realizability; see for instance [2], [9]. What we have proved is that *if* one knows that the observed pairs correspond to an analytic operator, then if any equation between this pairs is observed it follows that there is a realization as above. This *a priori* knowledge may be based on physical principles, for example. In principle, the theorem is fairly constructive, in the sense that from the observed relation E it should be possible to algorithmically obtain a realization. (We plan to devote some of our future efforts to the understanding of this issue.)

The proofs are based on a careful analysis of the concept of *observation space*, introduced in [8] and [5], and later rediscovered by many authors. The main technical lemma relates two different definitions of this space, one in terms of smooth controls and another in terms of piecewise constant ones; these two definitions are seen to coincide. One of them immediately relates to i/o equations, while the other is related to realizability through the notion of *observation algebras* and *observation fields*. These are the analogues of the corresponding discrete-time concepts studied in [11]; for differential equations they were first employed in [1].

Among future directions in this area, we are planning to concentrate on the study of the possibility of realizability by nonsingular polynomial systems as well as by "rational systems" (those for which q never vanishes). It is very possible that under extremely weak conditions every i/o operator might be realizable by some such systems, or at worst a system which can be naturally decomposed into a finite number of components of this type. Essentially this is the case for discrete-time in [11], where it is shown that realizability is equivalent to the existence of realizations whose state spaces are (Grothendieck) schemes which can be stratified in this fashion. In the present context, the treatment is complicated by the problem of understanding the meaning of differential equations over such spaces, as well the difficulties that arise when trying to prove existence and uniqueness results for differential equations on such spaces, i.e. integrability results for vector fields defined by formal derivation operators in the corresponding algebra of functions. The work [1] (and other papers by the same author) has already given preliminary results in that direction, however, and we plan to exploit these.

2 Some Technical Details

Let m be a fixed integer, and consider noncommuting variables $\eta_0, \ldots, \eta_m$. A *power series* in these variables is a formal expression

$$c = \langle c, \phi \rangle + \sum \langle c, \eta_\iota \rangle \eta_\iota$$

where the sum is over all possible sequences of indices

$$\iota = (i_1, \ldots, i_l),\ l \geq 0$$

with each $i_r \in \{0, \ldots, m\}$, including the empty sequence ε ($l = 0$), and where we denote

$$\eta_\iota := \eta_{i_1} \cdots \eta_{i_l},$$

and $\eta_\varepsilon := 1$. The coefficients $\langle c, \eta_\iota \rangle$ are real numbers. The set of all formal power series on $\eta_0, \ldots, \eta_m$ forms a real vector space under the coefficientwise operations $\langle rc_1 + c_2, \eta_i \rangle = r\langle c_1, \eta_i \rangle + \langle c_2, \eta_i \rangle$. We shall say that c is *convergent* if there exist $M, K > 0$ such that, for each sequence ι as above,

$$|\langle c, \eta_\iota \rangle| \leq KM^l l!.$$

For any fixed T, we let $\mathcal{U}_T$ to denote the set of all essentially bounded functions,

$$u : [0,\ T] \to \mathbb{R}^m$$

endowed with the L_1 topology.

Then for each T, each $u \in \mathcal{U}_T$, and each multiindex ι as above, we define inductively the functions $V_\iota = V_\iota[u] \in C[0,T]$ by $V_\varepsilon \equiv 1$ and

$$V_{i_1,\ldots,i_{k+1}}(t) = \int_0^t u_{i_1}(\tau) V_{i_2,\ldots,i_{k+1}}(\tau) d\tau,$$

where $u_i(\tau)$ is the i-th coordinate of $u(\tau)$ for $i = 1, \ldots, m$ and $u_0(\tau) \equiv 1$. It is easy to prove that each operator

$$\mathcal{U}_T \to C[0,T] : \ u \mapsto V_\iota[u]$$

is continuous. Further, if c is a convergent series and K, M are as above, then for $T < (Mm + M)^{-1}$, the series of functions

$$F_c[u](t) = F[u](t) = \sum \langle c, \eta_\iota \rangle V_\iota(t)$$

is absolutely and uniformly convergent for all $t \in [0,T]$ and all those $u \in \mathcal{U}_T$ such that $\sup |u_i(t)| \leq 1$ for all i; see [6], chapter III. Thus the operator F_c is also continuous on the subset of $\mathcal{U}_T$ satisfying this magnitude constraint. Further, c is in turn determined by F_c, in the sense that if $F_c = F_d$ for small enough T then $c = d$ (see [4]). If T and u are like this and u is of class C^{k-1}, then $y := F[u]$ is of class C^k; we call such a pair (u, y) a C^k *i/o pair* associated to c. It can also be proved that y is of class C^ω if u is of class C^ω.

Definition 2.1 The i/o map F_c satisfies an *algebraic i/o equation* if there exist an integer k and a nonzero polynomial E such that for any C^k i/o pair $w = (u,\ F[u])$,

$$E\left(w(t), w'(t), \ldots, w^{(k)}(t)\right) = 0 \ \text{ for any } t.$$

The i/o equation is called a *recursive equation* if it can be written as

$$a\left(u(t), \ldots, u^{(k)}(t)\right) y^{(k)}(t) = b\left(u(t), \ldots, u^{(k)}(t), y(t), \ldots, y^{(k-1)}(t)\right),$$

where a, b are polynomials and $a \not\equiv 0$. □

In realization theory, *observation spaces* play an important role. One may define the observation space in two different ways. The following is the first approach. For a power series

$$c = \langle c, \phi\rangle + \sum \langle c, \eta_\iota\rangle \eta_\iota$$

and monomial $\alpha = \eta_\iota$ we define $\alpha^{-1}c$ by

$$\langle \alpha^{-1}c, \eta_{i_1} \ldots \eta_{i_l}\rangle = \langle c, \eta_{j_1} \ldots \eta_{j_k}\eta_{i_1} \ldots \eta_{i_l}\rangle$$

for any $\alpha = \eta_{j_1} \ldots \eta_{j_k}$.

The *observation space* $\mathcal{F}_0$ is defined to be the space spanned by the $F_{\alpha^{-1}c}$'s over $\mathbb{R}$, i.e.,

$$\mathcal{F}_0 = \text{span}_{\mathbb{R}}\{F_{\alpha^{-1}c}\}_\alpha.$$

This space is in fact isomorphic to the space $\text{span}_{\mathbb{R}}\{\alpha^{-1}c\}_\alpha$.

The *observation algebra* $\mathcal{A}_0$ is defined to be the $\mathbb{R}$-algebra generated by $\mathcal{F}_0$, and the *observation field* $\mathcal{Q}_0$ is defined to be the quotient field of $\mathcal{A}_0$.

Note: It can be proved that $\mathcal{A}_0$ is an integral domain, thus $\mathcal{Q}_0$ is well defined.

Lemma 2.2 Let c be a convergent power series. Then

(a) F_c is realizable by a singular polynomial system if $\mathcal{Q}_0$ is a finitely generated field extension of $\mathbb{R}$.

(b) F_c is realizable by a non-singular polynomial system, (i.e., a singular polynomial system in which q is identically 1) if $\mathcal{A}_0$ is a finitely generated algebra. □

Now we discuss the second type of observation space. For the i/o map F_c, we define

$$G^{\mu_0\mu_1\cdots\mu_{k-1}}[u](t) := \frac{d^k}{d\tau^k}\Big|_{\tau=0_+} F_c[(u,t)(\omega,\tau)](t+\tau).$$

where $\omega(\tau) = \mu_0 + \mu_1\tau + \cdots + \mu_{k-1}\tau^{k-1}$ and $v = (u,t)(\omega,\tau)$ is the concatenated control defined by

$$v(s) = \begin{cases} u(s) & \text{for } 0 \le s \le t, \\ \omega(s) & \text{for } t < s \le t+\tau. \end{cases}$$

For $k = 0$ we define $G^\phi = F_c$.

The (second type of) *observation space* $\mathcal{F}_1$ is defined as follows:

$$\mathcal{F}_1 := \text{span}_{\mathbb{R}}\{G^{\mu_0\mu_1\cdots\mu_{k-1}} :\ k \ge 1,\ \mu_i \in \mathbb{R}, \}.$$

The corresponding *observation space* $\mathcal{A}_1$ is defined to be the $\mathbb{R}$-algebra generated by $\mathcal{F}_1$ and the *observation field* $\mathcal{Q}_1$ is the quotient field of $\mathcal{A}_1$. Again, $\mathcal{Q}_1$ is well defined since it can be proved that $\mathcal{A}_1$ is an integral domain.

Lemma 2.3 Suppose that c is a convergent power series. Then

(a) F_c satisfies an algebraic i/o equation $\Longrightarrow$ $\mathcal{Q}_1$ is a finitely generated field extension of $\mathbb{R}$.

(b) F_c satisfies a recursive i/o equation $\Longrightarrow$ $\mathcal{A}_1$ is a finitely generated $\mathbb{R}$-algebra. □

Lemma 2.2 shows that $\mathcal{F}_1$ is closely related to realizability of the i/o map while lemma 2.3 shows that F_2 is closely related to existence of an i/o equation. Thus the relation between $\mathcal{F}_0$ and $\mathcal{F}_1$ should give the desired relation between the realizability and the i/o equations. The following important lemma shows that the two spaces are indeed the same.

Lemma 2.4 The two observation spaces $\mathcal{F}_0$ and $\mathcal{F}_1$ are the same. □

To prove the lemma, we need the following fundamental formula:

$$\frac{d}{dt}F_c[u](t) = F_{\eta_0^{-1}c}[u](t) + u(t)F_{\eta_1^{-1}c}[u](t). \qquad (1)$$

Sketch of the proof of lemma 2.4:

By using equation (1), one can prove that

$$G^{\mu_0\mu_1\cdots\mu_{k-1}}[u](t)$$

is a polynomial of the variables $\mu_0, \mu_1, \ldots, \mu_{k-1}$ with coefficients $F_{\alpha^{-1}c}$. For example:

$$G^{\mu_0} = F_{\eta_0^{-1}c} + \mu_0 F_{\eta_1^{-1}c} \qquad (2)$$

$$\begin{aligned} G^{\mu_0\mu_1} &= F_{(\eta_0\eta_0)^{-1}c} \\ &+ \mu_0(F_{(\eta_1\eta_0)^{-1}c} + F_{(\eta_0\eta_1)^{-1}c}) \\ &+ \mu_0^2 F_{(\eta_1\eta_1)^{-1}c} + \mu_1 F_{\eta_1^{-1}c} \end{aligned} \qquad (3)$$

Therefore, we have

$$\mathcal{F}_1 \subseteq \mathcal{F}_0.$$

The other direction is less trivial. The following is the main idea of the proof.

From equation (2) and (3) one can see that

$$F_{\eta_0^{-1}c},\ F_{\eta_1^{-1}c} \in \mathcal{F}_1 \tag{4}$$

and

$$F_{(\eta_0\eta_0)^{-1}c},\ (F_{(\eta_1\eta_0)^{-1}c} + F_{(\eta_0\eta_1)^{-1}c}), \ldots, \in \mathcal{F}_1. \tag{5}$$

But $(2F_{(\eta_1\eta_0)^{-1}c} + F_{(\eta_0\eta_1)^{-1}c})$ is the coefficient of μ_1 in $G^{\mu_0\mu_1\mu_2}$. Therefore,

$$(2F_{(\eta_1\eta_0)^{-1}c} + F_{(\eta_0\eta_1)^{-1}c}) \in \mathcal{F}_1. \tag{6}$$

Combining 5 and 6, we see that both $F_{(\eta_1\eta_0)^{-1}c}$ and $F_{(\eta_0\eta_1)^{-1}c}$ are in $\mathcal{F}_1$

Main idea: Find enough many linear combinations of $\mathcal{F}_{\alpha^{-1}c}$'s which are in $\mathcal{F}_1$ to see that $\mathcal{F} \in \mathcal{F}_1$.

For example: Let

$$\alpha_j = \underbrace{\eta_0 \cdots \eta_0}_{n-j}\underbrace{\eta_1\eta_0 \cdots \eta_0}_{j}$$

and

$$X_j = F_{\alpha_j^{-1}c}.$$

Then for any $i \geq 0$, there exist a_{ij} such that $\sum_{j=1}^{n} a_{ij}X_j$ is the coefficient of μ_i in $G^{\mu_0 \cdots \mu_{n-1}}$. Therefore, there exist Y_i such that

$$\sum a_{ij}X_j = Y_i.$$

Let A be the matrix of n columns and infinitely may rows whose (i,j) is a_{ij}, i.e, $A = (a_{ij})$. Then it can be proved that A is of full (column) rank in the sense that there is no $X \in \mathbb{R}^n$ such that $AX = 0$. In fact,

$$A = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & C_2^1 & C_3^1 & \cdots & C_n^1 \\ 1 & C_3^2 & C_4^2 & \cdots & C_{n+1}^2 \\ 1 & C_4^3 & C_5^3 & \cdots & C_{n+2}^3 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & C_n^{n-1} & C_{n+1}^{n-1} & \cdots & C_{2n-1}^{n-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix}.$$

Let A_1 be the submatrix of A formed by the first n rows of A. Then one can see that A_1 is an $n \times n$ full rank matrix. Since

$$A_1X = \begin{pmatrix} Y_0 \\ \vdots \\ Y_{n-1} \end{pmatrix}, \quad \text{where } Y_i \in \mathcal{F}_1,$$

thus $X_i \in \mathcal{F}_1$, i.e., $F_{\alpha^{-1}c} \in \mathcal{F}_1$.

For general α's, it is hard to find out the explicit formula for the corresponding matrix A. But one can still prove that the matrix is full rank by using the fact that each convergent power series c is *uniquely* determined by F_c.

Now we can prove the main *theorem* stated in the introduction by using the above lemmas we obtained.

Proof of theorem 1:

Applying lemma 2.2, lemma 2.3 and lemma 2.4, we get the conclusion that the existence of an i/o equation implies realizability.

For the other direction, one can first prove that for those analytic i/o pair (u, y) corresponding to $q(x(t)) \not\equiv 0$, there is an i/o equation $E = 0$. And then by using the *regularity* property and continuity, one can prove that the equation is satisfied by every C^k i/o pair.

By using lemma 2.4 and part *(b)* in both lemma 2.2 and lemma 2.3, the following conclusion can be proved:

Theorem 2 *F_c can be realized by a non-singular polynomial system if F_c has a recursive i/o equation.* ∎

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FURTHER REMARKS ON PRESERVATION OF ACCESSIBILITY UNDER SAMPLING

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ABSTRACT*

A comparative presentation is given regarding accessibility results for both continuous-time and invertible discrete-time systems. By presenting these in a unified way, several results on in the discrete time case, as well as results on sampling, are shown to be simple consequences of the known continuous theory. A simple eigenvalue criterion, almost identical to that for linear systems, is given for preservation of accessibility for bilinear systems.

1. INTRODUCTION

This note presents an introduction to, and further results on, the preservation of nonlinear accessibility properties under sampling. Research into this general area was started in [SS]. Roughly, it deals with the question of deciding for which constant-rate sampling strategies (as appear in digital control) one does not loose controllability of a given nonlinear system.

A new, and simpler, proof is given of the main result in [SS], that strong accessibility guarantees sampled accessibility for all small enough frequencies. It is shown how this result is an easy consequence of the orbit theorems for continuous-time and discrete-time systems which are *invertible* (see below). A detailed comparison is given of these theorems.

Discrete-time control systems have been studied much less than their continuous counterparts, and their properties diverge considerably from those of the latter, due mainly to the possibilities of singularities; see for instance [SO]. The paper [JA] introduced the idea of studying *invertible* discrete nonlinear systems, and developed a realization theory which parallels much of the continuous time situation; further work along these lines was carried out in [FN], [NC], [SS], and related papers. Invertible systems are those for which transition maps, (one for each fixed control,) are all diffeomorphisms. Invertibility is of course a priori an extremely strong assumption in the context of general discrete time systems. However, for systems that result from the *sampling* of continuous time systems, this assumption is always satisfied. For invertible discrete-time systems, it is possible to give a characterization of accessibility that is analogous to that for continuous-time systems; see below. The paper [JN] showed that there is a yet another condition, involving Lie algebras, that is more closely related to the classical "accessibility rank condition". The proof there is based in expansion formulas from difference algebra.

We shall give below a new derivation of this condition, as a simple consequence of the corresponding "classical" facts about continuous-time systems. The proof is of independent interest, in that it introduces a rather general method for reducing some questions about invertible discrete-time systems to their already studied continuous-time analogues. Motivated by an important observation in [NC], we show that in the context of sampling there is an even simpler condition that characterizes accessibility for small enough δ. The proof depends on some results about parametrized vector fields.

In the case of bilinear systems $\dot{x} = (A+u_1B_1+\cdots+u_rB_r)x$, it is shown that if the strong accessibility rank condition holds at all $x\neq0$ and if $\delta>0$ is such that $\delta(a+b-c-d)\neq 2k\pi i$ for all eigenvalues a,b,c,d of A and all nonzero integers k, then the system is also accessible with controls sampled at frequency $1/\delta$. This is analogous to the classical Kalman-Bertram result for linear systems.

Some proofs are omitted for lack of space. A full length version of the paper can be obtained from the author.

2. SYSTEMS

2.1. Continuous-time systems

The continuous-time systems Σ to be considered are described by equations

$$\dot{\xi}(t) = P(\xi(t),\mu(t)) \tag{1}$$

where states $\xi(t)$ belong to a smooth (i.e., C^∞) Hausdorff second countable connected n-dimensional manifold **M**, and controls μ take values in a subset **U** of a manifold $\mathcal{U}$. Here $P_u := P(\cdot,u)$ is a smooth complete vector field for each $u\in$**U**. (Completeness is a strong and somewhat unrealistic assumption. Many sampling results can be however generalized, provided that appropiate care be taken in the choice of sampling times; see for instance the treatment in [SO1].) Further we assume that P extends to a smooth map **M**$\times\mathcal{U}\to$**TM**, int(**U**) is connected, and for each $u\in$**U** there is a smooth path $\gamma: [0,1]\to\mathcal{U}$ with $\gamma(0)=u$, $g([0,1])\subseteq$**U**, and such that $\gamma(t)$ is in int(**U**) for almost all t. In particular, then, **U** must be path-connected and it must satisfy **U**$\subseteq$clos(int(**U**)).

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An *analytic* system is one for which $\mathbf{M}$ is real-analytic, and all P_u are analytic. A *control-linear* system is one for which $\mathbf{U} = \mathbb{R}^m$ and $P(\xi,u)$ is linear in u: $P(\xi,u) = f(\xi) + \Sigma u_i g_i(\xi)$, where the g_i, $i = 1,\cdots,m$ are appropiate vector fields. A *control (function)* is a μ: $[0,T) \rightarrow \mathbf{U}$ which is measurable and essentially bounded; $T = |\mu|$ is the *length* of μ. For references on results to be used without proof, see for instance [SJ], [HK], [KL], [SU2], and [KR].

A rather heavy amount of notations and definitions is unavoidable. We first give some differential-geometric terminology which generalizes that in the standard literature. This follows [KR], with one exception as noted. The Lie algebra of all smooth vector fields on $\mathbf{M}$ is $\Xi(\mathbf{M})$. We denote by $X(\xi)$, instead of X_ξ, the value of $X\in\Xi(\mathbf{M})$ at $\xi\in\mathbf{M}$. A (singular) *distribution* D on $\mathbf{M}$ is a linear subspace (not necesarily a $C^\infty(\mathbf{M})$-submodule as in [KR]) of $\Xi(\mathbf{M})$. For each $\xi\in\mathbf{M}$, $D(\xi) := \{Z(\xi), Z\in D\}$. The *singular subbundle* of $T\mathbf{M}$ associated to D is the union of the subspaces $D(\xi)$, ξ in $\mathbf{M}$. The *completion* $\hat{D}$ of D is the singular distribution consisting of all vector fields $X\in\Xi(\mathbf{M})$ such that $X(\xi)\in D(\xi)$ for all $\xi\in\mathbf{M}$. The *rank* of D at ξ is the dimension of $D(\xi)$ –equivalently, the dimension of $\hat{D}(\xi)$. (Thus the constant rank case corresponds to the usual notion of distribution in differential geometry.) An *integral submanifold* N of D is a connected (necessarily second countable) immersed submanifold of $\mathbf{M}$ such that $T_\xi N = D(\xi)$ for each ξ in N. The distribution D is *integrable* iff it induces a (singular) foliation, i.e., there is a partition of $\mathbf{M}$ into maximal integral manifolds ("leaves") of D.

If $\Phi \subseteq \Xi(\mathbf{M})$, $\{\Phi\}_{LIN}$ denotes the linear subspace (distribution) generated by Φ and $\{\Phi\}_{LA}$ is the Lie subalgebra generated by Φ. Thus $\{\Phi\}_{LA}$ consists of all possible finite linear combinations of iterated Lie brackets formed from the vector fields in Φ.

The state $x(\tau)$ at time $\tau\in[0,T]$ corresponding to solving equation (1) with $x(0)=\xi$ and control μ, is denoted by $\phi[\tau,\xi,\mu]$; when $\tau = |\mu|$ this is denoted just by $\phi[\xi,\mu]$. The flow generated by a vector field X is denoted by $\{e^{tX}, t\in\mathbb{R}\}$. Thus if μ is constantly equal to u on $[0,T)$ and $X = P_u$, then $e^{tX}(\xi) = \phi[\xi,\mu]$ for all ξ.

The state ζ is *reachable from* ξ *(in time* $|\mu|$*)* iff there is a control μ such that $\zeta = \phi[\xi,\mu]$. Reachability being too difficult to study, it is customary to consider instead accessibility, the equivalence relation generated by reachability. Thus, ζ is *accessible from* ξ iff there is a sequence of states $\xi_o=\xi$, ξ_1, $\cdots$, $\xi_k=\zeta$, for some $k\geq0$, such that for each i, either ξ_i is reachable from ξ_{i+1} or ξ_{i+1} is reachable from ξ_i. The set of states accessible from x is A(x). The set of states accessible "in zero time" from ξ is $A^o(\xi)$: $\zeta\in A^o(\xi)$ iff there are reals $t_1,\cdots,t_k$, $\Sigma t_i=0$, and a sequence of states $\xi_o=\xi$, ξ_1, $\cdots$, $\xi_k=\zeta$, such that for each i either ξ_i is reachable from ξ_{i+1} in time $-t_i$ or ξ_{i+1} is reachable from ξ_i in time t_i. We shall say that the system Σ is *accessible at* ξ [resp., *strongly accessible at* ξ] iff ξ is in the interior of $A(\xi)$ [resp., of $A^o(\xi)$].

If Φ is any subset of $\Xi(\mathbf{M})$ (consisting of complete vector fields), we may consider the group of diffeomorphisms $\exp(\Phi)$ = group generated by all $\{e^{tX}$, $X\in\Phi$ and $t\in\mathbb{R}\}$. Let $O_\Phi(\xi)$ be the orbit of ξ under $\exp(\Phi)$.

We introduce now some notation that will be especially useful in comparing the discrete-time and continuous-time results to be given later. If π: $\mathbf{M}\rightarrow\mathbf{M}$ is a diffeomorphism, we denote by Ad_π the (linear) operator $\Xi(\mathbf{M})\rightarrow\Xi(\mathbf{M})$ corresponding to conjugation by π, more precisely: $Ad_\pi(X)(\xi) := (\pi^{-1})_*(X(\pi(\xi)))$ for X in $\Xi(\mathbf{M})$ and ξ in $\mathbf{M}$. In this equation, $(\cdot)_*$ denotes the differential of π^{-1} at the point $\pi(\xi)$. When $\pi = e^X$ for a (complete) vector field X, we denote Ad_π simply as Ad_X. If Γ is a group of diffeomorphisms on $\mathbf{M}$, and Φ is a subset of $\Xi(\mathbf{M})$, we introduce the distribution $Ad_\Gamma(\Phi) := \{Ad_\pi(X), \pi\in\Gamma$ and $X\in\Phi\}_{LIN}$. In particular, we denote $Ad(\Phi,\Phi) := Ad_{\exp(\Phi)}(\Phi)$. By a well-known theorem of Sussmann ([SU1]), $Ad(\Phi,\Phi)$ is integrable, and its leaves are the orbits $O_\Phi(\xi)$. Further, if Σ is a system and $\Phi = \Phi_\Sigma$, then for each ξ, $O_\Phi(\xi) = A(\xi)$.

Proposition 1: The distribution $Ad(\Phi,\Phi)$ is integrable, and the leaves of the corresponding foliation are the sets $A(\xi) = O_\Phi(\xi)$.

Corollary 2: Accessibility at ξ is equivalent to $Ad(\Phi,\Phi)$ having rank n at ξ.

There is an analogous characterization of strong accessibility, as follows. Consider the distribution generated by the set of differences of the generators given for $Ad(\Phi,\Phi)$, $Ad_o(\Phi,\Phi) := \{Ad_\pi(X)-Ad_\rho(Y), \pi,\rho\in\exp(\Phi), X,Y\in\Phi\}_{LIN}$. Then, the distribution $Ad_o(\Phi,\Phi)$ is integrable, the corresponding leaves are the sets $A^o(\xi)$, and strong accessibility is equivalent to this distribution having full rank. (This 'classical' fact, as well as the previous result, can be obtained as immediate consequences of the orbit theorem in [SS]. See also the reference [KL] for a proof in the strong accessibility case.)

Yet another result that follows from the orbit theorem mentioned in the previous paragraph is an alternative characterization of strong accessibility in terms of a rather different distribution. This other distribution arises from an alternative approach to the construction of the differentiable structure on orbits than that used in [SU1]. While that construction is based on generating tangent spaces by variations in time, the alternative, when dealing with systems satisfying the axioms we have given, is to introduce variations in the control space.

The argument, when applying the result in the appendix of [SS] with an appropiate choice of data, is roughly as follows. Let ξ be in **M**. Consider all possible ways of going from states ζ into the state ξ using positive or negative time piecewise constant trajectories $e^{t_1X_1}\circ\cdots\circ e^{t_nX_n}(\zeta)$, with each X_i of the form $P_{u'}$, $u=u_i$. Now consider the infinitesimal directions at ξ induced by perturbations of the controls u_i. These directions, for all possible such sequences of controls, can be used to define a submanifold structure for $A(\xi)$, a structure which is in principle *different* from that obtained when using time variations. This manifold may be disconnected in its own topology; its connected component at ξ is $A^o(\xi)$ (see remark A.13 in the above reference). The above directions generate the tangent space to $A^o(\xi)$, or equivalently $A(\xi)$, at ξ. More precisely, one obtains the distribution introduced below.

First we describe a general construction, which will be used again in a different context later. Assume given a smooth map Ω: $\mathbf{M}X\mathcal{U}\longrightarrow\mathbf{M}$ such that $\Omega_u = \Omega(\cdot,u)$ is a diffeomorphism for each u belonging to a subset U of $\mathcal{U}$ that satisfies the conditions we imposed earlier. For each (u,ν) in the tangent bundle TU (meaning, to be precise, $u\in\mathbf{U}$ and tangent vector ν in $T_u\mathcal{U}$,) consider the vector field $Q_{u,\nu}\in\Xi(\mathbf{M})$ defined by:

$$Q_{u,\nu}(\xi):=\frac{\partial\Omega_v(\Omega_u^{-1}(\xi))}{\partial v}\Big|_{v:=u}(\nu). \tag{2}$$

It is clear how to compute the above in local coordinates (product of a Jacobian matrix by a vector). A coordinate-free interpretation is as follows. Let α: $\mathbf{U}\longrightarrow\mathbf{M}$ be the map that sends $v\in\mathbf{U}$ into $\Omega_v(\Omega_u^{-1}(\xi))$. For any smooth map f: $\mathbf{M}\longrightarrow\mathcal{R}$, consider the composition $\beta:= f\circ\alpha$. Then $Q_{u,\nu}(\xi)(f)$ is the evaluation $\nu(\beta)$, where we are interpreting ν as a differential operator on germs of functions at u. With this definition, it is clear that $Q_{u,\nu}(\xi)$ is again a differential operator, and so defines a vector at ξ. From the coordinate description it follows that this is not only smooth on ξ but in fact smooth as a function of $(\xi,u,\nu)\in\mathbf{M}X\mathrm{TU}$. Similarly, for such u, ν, let $Q'_{u,\nu}$ be given by $Q'_{u,\nu}(\xi):=\dfrac{\partial\Omega_v^{-1}(\Omega_u(\xi))}{\partial v}\Big|_{v:=u}(\nu)$. We apply this general construction to systems Σ as follows. For any fixed $t\in\mathcal{R}$, consider the mapping $\Omega(\xi,u):= e^{tP_u}(\xi)$. Let Φ_t^* be the set of vector fields of the form $Q_{u,\nu}$ and $Q'_{u,\nu}$ for all possible (u,ν) in **TU**, and let Φ^* be the union of all Φ_t, $t\in\mathcal{R}$. As before, let $\Gamma = \exp(\Phi)$, where $\Phi = \Phi_\Sigma$. Consider the distribution $Ad_\Gamma(\Phi^*)$. Then,

Proposition 3: The distribution $Ad_\Gamma(\Phi^*)$ is integrable, and the leaves of the corresponding foliation are the sets $A^o(\xi)$.

Corollary 4: Strong accessibility at ξ is equivalent to $Ad_\Gamma(\Phi^*)$ having rank n at ξ.

It will be of interest to consider a couple of other distributions related to the accessibility property. These are *Lie algebras*, and are spanned by far fewer generators than the above. For any subset Φ of $\Xi(\mathbf{M})$, let $L(\Phi) := \{\Phi\}_{LA}$, and $L_o(\Phi)$:= ideal of L generated by $\{X-Y$, for X,Y in $\Phi\}$. (We omit Φ if clear from the context.) If Σ is control-linear, and $\Phi = \Phi_\Sigma$, it is well-known that $L = \{f,g_1,\cdots,g_m\}_{LA}$, and $L_o = \{ad_f^j(g_i),\ j\geq 0,\ i = 1,\cdots,m\}_{LA}$, where $ad_f(X) = [f,X]$. The *accessibility rank condition* at $x\in\mathbf{M}$ is the condition $\dim L(x) = n$. The *strong accessibility rank condition* at x is the condition $\dim L_o(x) = n$.

It is a standard result in nonlinear control that the accessibility rank condition at ξ implies accessibility at ξ, and that the converse holds if all vector fields in Φ are analytic (e.g. for analytic systems), or for the more general smooth case under extra assumptions, like the Frobenious condition of constant rank on L, or alternatively finite dimensionality of L. The strong condition is related to strong accessibility in the same manner.

Proposition 5: The [strong] accessibility rank condition at ξ implies [strong] accessibility at ξ. If all vector fields in Φ are analytic, the converses also hold.

2.2. Discrete-time systems

We shall mainly consider discrete-time systems that appear through sampling, but some facts can be stated in a somewhat more general framework. An *invertible discrete-time system* Σ is a system defined by equations

$$\xi(t+1) = P(\xi(t),\mu(t)) \quad (t \text{ integer}) \tag{3}$$

where states $\xi(t)$ belong to a smooth manifold **M** as above, controls take values $u=\mu(t)$ in a set **U** as in section 2.1, the map P extends to a smooth map $\mathbf{M}X\mathcal{U}\longrightarrow\mathbf{M}$, and $P_u := P(\cdot,u)$ is a diffeomorphism for each $u\in\mathbf{U}$ (invertibility). An *analytic* system is one for which **M** is real-analytic, and each P_u is analytic. We define reachability, accessibility, and strong accessibility just as for continuous-time systems, and use again the notations $A(\xi)$, $A^o(\xi)$.

For each invertible system Σ there is a foliation whose leaves correspond to the sets $A^o(\xi)$. The sets $A(\xi)$ are all countable unions of leaves, and for any $\zeta\in A(\xi)$ there is a diffeomorphism of **M** mapping $A^o(\xi)$ onto $A^o(\zeta)$. It follows by a category argument that accessibility and strong accessibility coincide. The foliation is constructed in exactly the same way as its continuous time analog $\mathrm{Ad}_\Gamma(\Phi^*)$. Again the tangent space of $A^o(\xi)$ (equivalently, of $A(\xi)$,) is generated by perturbations of control values.

Let Ω in the discussion in the previous section be the discrete time transition map P. Introduce the vector fields $Q_{u,\nu}$ and $Q'_{u,\nu}$ as there. Let Γ be now the group of diffeomorphims generated by $\{P_u, u\in U\}$, and let Φ be the set of vector fields of the form $Q_{u,\nu}$ and $Q'_{u,\nu}$ for all possible (u,ν) in TU. Consider the corresponding distribution $\mathrm{Ad}_\Gamma(\Phi)$. Then, *Proposition 3 and Corollary 4 are valid with the present interpretations:*

Proposition 6: The distribution $\mathrm{Ad}_\Gamma(\Phi)$ is integrable, and the leaves of the corresponding foliation are the sets $A^o(\xi)$.

Corollary 7: (Strong) accessibility at ξ is equivalent to $\mathrm{Ad}_\Gamma(\Phi)$ having rank n at ξ.

When $U = \mathbb{R}^m$, –or more generally for parallelizable U,– one has a somewhat simpler description of $\mathrm{Ad}_\Gamma(\Phi)$. Let $\{e_1,\cdots,e_m\}$ be vector fields on U such that $\{e_1(u),\cdots,e_m(u)\}$ is a basis of T_uU for each u. For each $u\in U$ and each $i = 1,\cdots,m$, let $Q_{u,i}$ be $Q_{u,\nu}$ with $\nu:= e_i(u)$. Similarly for the $Q'_{u,\nu}$. Let Φ_1 be the set of vector fields thus obtained. Since $Q_{u,\nu}$ (as well as $Q'_{u,\nu}$) is linear in $\nu\in T_uU$ for each fixed u, it follows that $\{\Phi\}_{LIN} = \{\Phi_1\}_{LIN}$. Thus $\mathrm{Ad}_\Gamma(\Phi) = \mathrm{Ad}_\Gamma(\Phi_1)$.

The distribution $D = \mathrm{Ad}_\Gamma(\Phi)$ is such that $D(\xi) = T_\xi A^o(\xi)$, but other naturally defined larger distributions (with the same associated "singular bundle") may still satisfy this property. In particular, consider the vector fileds defined for each positive k and each (u,ν) as above via:

$$\tilde{Q}_{k,u,\nu}(\xi) := \frac{\partial P^k_v(P^{-k}_u(\xi))}{\partial v}\Big|_{v:=u}(\nu). \qquad (4)$$

(For k=1, this is the vector field $Q_{u,\nu}$ introduced earlier.)

Similarly, one can define vector fields $\tilde{Q}'_{k,u,\nu}$ by interchanging the inverses. Let $\tilde{\Phi}$ be the set of all possible $\tilde{Q}_{k,u,\nu}$ and $\tilde{Q}'_{k,u,\nu}$; thus $\Phi \subseteq \tilde{\Phi}$. Consider the distribution $\tilde{D} := \mathrm{Ad}_\Gamma(\tilde{\Phi})$, where Γ is as before. So $\tilde{D}(\xi)$ contains $T_\xi A^o(\xi)$. It is easy to see by a continuity argument that they are in fact equal.

3. Sampling

Fix a continuous-time system Σ as in section 2.1. Let $\sigma=(s_1,\cdots,s_r)$ be a sequence of positive real numbers, with $T:=\sum s_i$, and let $u:=(u_1,\cdots,u_r)$ be a sequence of elements in U. Then u_σ is the control function μ of length T defined as follows by $\mu(t):= u_i$ if $t\in[s_0+\cdots+s_{i-1}, s_0+\cdots+s_i)$, $i=1,\cdots,r$ (denoting $s_0:=0$). If δ is a positive real and $u:=(u_1,\cdots,u_r) \in U$, u^r_δ is the δ-*sampled* control u_σ, where $\sigma:= (\delta,\cdots,\delta)$ (r times).

If $\zeta = \phi[\xi,\mu]$ and $\mu = u^r_\delta$ is δ-sampled, we say that ζ is δ-*reachable* from ξ (in r steps). The δ-accessibility relation is the equivalence relation generated by δ-reachability, and $A_\delta(\xi)$ is the set of states δ-accessible from ξ. Finally, $A^o_\delta(\xi)$ is the set of states δ-accessible in zero time.

For any given $\delta>0$ let Σ_δ be the invertible discrete-time system obtained when δ-sampled controls are used in equation (1). The notions of accessibility introduced in the previous paragraph are then consistent with those given earlier for discrete-time systems. We say that Σ is δ-*accessible at* ξ iff Σ_δ is accessible at ξ, i.e. ξ is in the interior of $A_\delta(\xi)$. By the remarks in section 2.2, this is equivalent to the corresponding "strong δ-accessibility" notion that $\xi\in\mathrm{int}(A^o_\delta(\xi))$.

Before stating the basic result on sampling, we give a (very) easy lemma on matrices. This will be applied in a couple of places later, and also gives as a corollary the result in the appendix of [SO2]. If A is a set of real matrices all of size $\rho\times\sigma$, we let $S(A)$ denote the subspace of $\mathbb{R}^\rho$ generated by the columns of all the matrices in A. For any real δ and any k, $\delta\mathbf{Z}^k$ is the lattice of $\mathbb{R}^k$ consisting of all $t=(t_1,\cdots,t_k)$ with t_i/δ = integer for each i. If $A(t) = A(t_1,\cdots,t_k)$ is an $\rho\times\sigma$-matrix of smooth functions of t, and if α is a k-vector of nonnegative integers adding to r, we denote the (componentwise) α-derivative of A evaluated at t, by $A^\alpha(t)$.

Lemma 8: Assume that A(t) is as above, and consider the following statements:

(1) $S(\{A(t),\ t\in\mathbb{R}^k\}) = \mathbb{R}^\rho$,

(2) For some $\Delta>0$, $S(\{A(t), t\in\delta\mathbf{Z}^k\})=\mathbb{R}^\rho$, all $0<\delta<\Delta$,

(3) $S(\{A^\alpha(0),\ \alpha \text{ multiindex}\}) = \mathbb{R}^\rho$.

Then, (1) is equivalent to (2), and (3) implies both. If A is analytic in t, all are equivalent.

Theorem 9: If Σ is strongly accessible at ξ, then there is a $\Delta>0$ such that Σ is δ-accessible at ξ for all $0<\delta<\Delta$. Conversely, if Σ is δ-accessible at ξ for any $\delta>0$ then Σ is strongly accessible at ξ.

The proof is easy from the application of the results in section 2.2 to the case of systems of the form Σ_δ. It is enough to show that the distribution $\bar{D}$ has full rank at ξ, and this follows from the above lemma applied to the generators of the distributions introduced earlier.

4. A discrete-time criterion

Let Σ be a discrete-time system as in section 2.2. Fix an arbitrary control value $u_0 \in \mathbf{U}$. We denote this simply as 0. Let π: $\mathbf{M} \to \mathbf{M}$ be the diffeomorphism $P(\cdot,0)$, and let Γ be the cyclic group generated by π. Assume that ζ is in $A^O(\xi)$, that is, there are control values $u_1, \cdots, u_\ell$ and integers $\epsilon_i = \pm 1$ adding to zero, such that $\zeta = P_{u_1}^{\epsilon_1} \circ \cdots \circ P_{u_\ell}^{\epsilon_\ell}(\xi)$. For each nonnegative integer k and each $u \in \mathbf{U}$, let $P^{(k)}_{u,\epsilon}$ be the diffeomorphism defined by the composition $\pi^{-k} \circ P_u \circ \pi^{k-1}$ when $\epsilon=1$ and $\pi^{k-1} \circ P_u \circ \pi^{-k}$ when $\epsilon=-1$. Padding with suitable powers of π (positive and negative), there is then a sequence of states $\xi_0=\xi$, $\cdots$, $\xi_\ell=\zeta$ such that for each i, there are u, k, and ϵ such that either $\xi_{i+1} = P^{(k)}_{u,\epsilon}(\xi_i)$ or $\xi_i = P^{(k)}_{u,\epsilon}(\xi_{i+1})$.

Fix any of these i, and consider the corresponding u. Assume that the first case holds (otherwise the roles of ξ_i and ξ_{i+1} are interchanged). Because of the hypothesis on $\mathbf{U}$, there is a smooth curve γ: $[0,1] \to U$ such that $\gamma(0)=0$, $\gamma(1)=u$, and $\gamma(t)$ is in $\mathbf{U}$ for all t. Thus the following situation holds for each such i: there is a state y (namely, $\pi^{k-1}(\xi)$ when $\epsilon=1$, or $\pi^{-k}(\xi)$ if $\epsilon=-1$,) such that $\xi(t) := \pi^j \circ P_{\gamma(t)}(y)$ satisfies $\xi(0)=\xi_i$, $\xi(1)=\xi_{i+1}$, where j is an integer ($j=-k$ when $\epsilon=1$, $j=k-1$ when $\epsilon=-1$). Note that $\xi(\cdot)$ is a smooth curve, and

$$\dot{\xi}(t) = Ad^j_\pi(Q_{\gamma(t),\nu(t)})\,(\xi(t))\,, \tag{5}$$

where we denote $\nu(t) := \dot{\gamma}(t)$ (element of $T_{\gamma(t)}\mathbf{U}$). We now define a distribution in the following way. Let $L := \{Ad^j_\pi(Q_{u,\nu}), u\in\mathbf{U}, j\in\mathbf{Z}, \nu\in T_u\mathbf{U}\}_{LA}$. The generators of L are in particular in the distribution $Ad_\Gamma(\Phi)$ that appears in Proposition 6, and thus are tangent to the sets $A^O(\xi)$. It follows that $L(\xi) \subseteq T_\xi A^O(\xi)$ for each ξ, so that

Corollary 10: If L has rank n at ξ then Σ is strongly accessible at ξ.

We now give a simple proof of the result in [JN] that for analytic systems the converse also holds:

Proposition 11: If Σ is analytic and is strongly accessible at ξ then L has rank n at ξ.

Proof: Let Φ be the set of the generators given above for L. Note that these are all analytic vector fields. We shall prove that the orbit $O_\Phi(\xi)$ contains ξ in its interior. By corollary 2, and proposition 5, it will follow that L ($=\mathcal{L}$ in that context) must have rank n at ξ. By assumption, there is an open neighborhood W of ξ contained in $A^O(\xi)$. We claim that $W \subseteq O_\Phi(\xi)$. Pick any $\zeta \in W$. By the preceeding discussion, we may assume without loss of generality that there is a smooth path $\xi(\cdot)$ with $\xi(0)=\xi$ and $\xi(1)=\zeta$, whose derivative satisfies equation (5). We now define a *continuous-time* system Σ' as follows. The control space is the tangent bundle $T\mathbf{U}$, the state space is again $\mathbf{M}$, and, (with the given j,) $P(\xi,u,\nu) := Ad^j_\pi(Q_{u,\nu})(\xi)$. Thus $\xi(\cdot)$ is a trajectory of Σ'. It follows that ζ is in $A'(\xi)$ (the accessible class relative to Σ'), which is contained in $O_\Phi(\xi)$ because $\Phi_{\Sigma'} \subseteq \Phi$. This completes the proof.∎

An even smaller set of generators, not sufficient in general even in the analytic case, will be very useful in the context of sampling. For simplicity, we describe these only in the case of paralellizable U. For each $i = 1,\cdots,m$ and each positive integer j let

$$b_{i,j}(\xi) := \frac{\partial P^j_v(\pi^{-j}(\xi))}{\partial v}\Big|_{v:=0}(e_i(0))\,,$$

where u=0 is the element of $\mathbf{U}$ which we fixed at the start of this section. (These are particular cases of the vector fields $\tilde{Q}_{k,u,\nu}$ introduced earlier.) We denote $b_{i,1}$ simply as b_i. This is the same as $Q_{0,i}$ in formula (2), with $\nu=e_i(0)$. In particular, $Ad^j_\pi(b_i)$ is in L for each $j\in\mathbf{Z}$ and $i = 1,\cdots,m$. Calculating with coordinates, we verify the formula $b_{i,j+1} = Ad_\pi(b_{i,j}) + b_i$, $i = 1,\cdots,m$, $j\geq 0$. By induction, we conclude that $\{b_{i,j}, i = 1,\cdots,m, j\geq 1\}_{LA} = \{b_1, \cdots, b_m, Ad_\pi(b_1), \cdots, Ad_\pi(b_m), Ad^2_\pi(b_1), \cdots\}_{LA}$. Let $\mathcal{L}_\delta$ be this distribution. Since the right-hand side is included in L, we conclude that *if* $\mathcal{L}_\delta$ *has rank n at* ξ *then* Σ *is accessible at* ξ.

5. A result on parametrized vector fields

By a *parametrized vector field* (for short, *p.v.f.*) we shall mean a function $\underline{X} : \mathcal{R} \to \Xi(\mathbf{M})$ such that $X_\tau := \underline{X}(\tau)$ is complete for each τ and $X_\tau(\xi)$ depends smoothly on $(\tau,\xi)\in\mathcal{R}\times\mathbf{M}$. If $\underline{X}$ and $\underline{Y}$ are like this, $\underline{Z} := [\underline{X},\underline{Y}]$ is by definition the parametrized vector field with $Z_\tau := [X_\tau,Y_\tau]$ for each τ. This defines a Lie algebra structure on the set of all parametrized vector fields. If $\underline{X}$ is a p.v.f., we may consider the new p.v.f. $\underline{X}'$ obtained by τ-derivation, as well as higher order derivatives. If $\underline{X}$ is a p.v.f., we denote by ${}^k\underline{X}$ the new p.v.f. with ${}^kX_\tau = X_{k\tau}$. A *shift-invariant family* of parametrized vector fields is a set **F** of such $\underline{X}$, with the property: *If* $\underline{X}$ *is in* **F** *and* k *is a positive integer, then* ${}^k\underline{X}$ *is again in* **F**. For any family of p.v.f.'s **F**, $\{\mathbf{F}\}_{LA}$ is the Lie algebra (in the sense of the previous paragraph) generated by **F**. Note that $\{\mathbf{F}\}_{LA}$ is again a shift-invariant family if **F** is. We consider also the new family, for each **F**, $\mathbf{F}^\infty := \{\underline{X}^{(N)}, N\geq 0, \underline{X}\in\mathbf{F}\}$. If **F** is shift-invariant, then $\mathbf{F}^\infty$ is too. Finally, let $\mathbf{F}_o^\infty$ be the set of all derivatives at 0 of the p.v.f.'s in **F**, $\mathbf{F}_o^\infty := \{ X_o^{(N)}, N\geq 0, \underline{X}\in\mathbf{F} \} \subseteq \Xi(\mathbf{M})$.

Lemma 12: For any shift-invariant family **F**, $\{\mathbf{F}_{LA}\}_o^\infty = \{\mathbf{F}_o^\infty\}_{LA}$.

(Note that both sets represent Lie algebras of vector fields, *not* of p.v.f.'s.) For any family **F** and each $\delta>0$ we denote $\mathbf{F}_\delta := \{X_\delta, \underline{X}\in\mathbf{F}\}_{LA} \subseteq \Xi(\mathbf{M})$. Note that $\{\mathbf{F}_{LA}\}_\delta = \{\mathbf{F}_\delta\}_{LA}$.

Lemma 13: If **F** is a shift-invariant family of p.v.f.'s such that the distribution $\mathbf{F}_o^\infty$ has rank n at ξ then there is a $\Delta>0$ such that $\mathbf{F}_\delta$ has rank n at ξ for each $0<\delta<\Delta$.

The proof uses the above and lemma 8.

Corollary 14: If **F** is a shift-invariant family of p.v.f.'s such that $\{\mathbf{F}_o^\infty\}_{LA}$ has rank n at ξ then there is a $\Delta>0$ such that $\{\mathbf{F}_\delta\}_{LA}$ has rank n at ξ for each $0<\delta<\Delta$.

We apply the above to the following situation in sampling. For simplicity, we treat only the case of parallelizable U. Fix an element u_o, say "0", in **U**, and let w_i be $e_i(0)\in T_o\mathbf{U}$. Finally, fix a continuous-time system Σ. For each $i = 1,\cdots,m$, positive integer k, and $t\in\mathbb{R}$, consider the p.v.f. $\underline{X}^{[i,k]}$ defined by, for each τ,

$$X^{[i,k]}(\xi) := \frac{\partial e^{k\tau P_v} e^{-k\tau P_o}(\xi)}{\partial v}\Big|_{v=0}(w_i).$$

This is the same as $Q_{u,\nu}$ in equation (2), with $\Omega(\xi,v) := e^{k\tau P_v}$ and $u=0$, $\nu=w_i$. Let **F** be the family consisting of all such p.v.f.'s; **F** is shift-invariant because ${}^{\ell}\underline{X}^{[i,k]} = \underline{X}^{[i,k\ell]}$. Further, for each fixed $\delta>0$, $\mathbf{F}_\delta$ is nothing else than the set of all elements $b_{i,j}$ (with respect to the system Σ_δ). It follows that Σ is δ-accessible if $\{\mathbf{F}_\delta\}_{LA}$ has rank n at ξ, or, from corollary 14,

Corollary 15: If $\{\mathbf{F}_o^\infty\}_{LA}$ has rank n at ξ then there is a $\Delta>0$ such that Σ is δ-accessible at ξ whenever $0<\delta<\Delta$.

Now let Σ be a control-linear system. The critical observation at this point, due to [NC], is that $\{\mathbf{F}_o^\infty\}_{LA}$ is precisely the algebra L_o that appears in the accessibility rank condition. This follows from a formula in Lie theory (see for instance [GO]): with $P_v = f + \Sigma v_i g_i$, $v\in\mathbb{R}^m$, $P_o = f$, and the obvious choice of e_i's, one verifies from the theory of Lie series (or directly by a simple computation) that

$$(X^{[i,k]})_o^{(N)} = k^N ad_f^{N-1}(g_i). \tag{6}$$

Corollary 16: If Σ satisfies the strong accessibility rank condition at ξ then there is a $\Delta>0$ such that for each $0<\delta<\Delta$, the distribution L_δ has rank n at ξ. In particular, Σ is δ-accessible for all such δ.

Note that the last conclusion is a very particular case of a fact that we had already proved above (c.f. theorem 9). The first conclusion is rather interesting, however. Based on the discrete-time theory, there is no reason to expect the distribution L_δ to have full rank, even if Σ is indeed δ-accessible. But the corollary says that when dealing with sampling problems, *at least for small enough* δ this will indeed be true.

6. Bilinear systems

As an illustration of the previous discussion, we consider now the case of continuous-time systems Σ for which the Lie algebra L is finite dimensional. Let Σ be such a system. For any fixed $\delta>0$, consider the distribution L_δ. Let A be the linear operator ad_f on the (finite-dimensional) vector space L. Then the linear operator $A^{(\delta)} := \delta\Sigma_{N=1}^\infty (\delta A)^{N-1}/N!$ is well-defined, and $b_i = A^{(\delta)}g_i$. Note that formally $A^{(\delta)}$ is $(e^{\delta A}-I)A^{-1}$, but A here is never invertible. With $\pi = e^{\delta f}$, the Baker-Campbell-Hausdorff formula gives that $Ad_\pi(b_i) = e^{\delta A}b_i$. Thus Σ is δ-accessible at ξ if $L_\delta = \{A^{(\delta)}g_1, \cdots, A^{(\delta)}g_m, e^{\delta A}A^{(\delta)}g_1, \cdots, e^{\delta A}A^{(\delta)}g_m, e^{2\delta A}A^{(\delta)}g_1,\cdots\}_{LA}$ has rank n at ξ. Consider now the *linear* continuous-time system with state space L and m controls, defined by the pair (A,B), where B has columns $g_1,\cdots,g_m$. This system may not be controllable; let Σ^ℓ be its restriction to the span of $\{A^j g_i,\ i = 1,\cdots,m,\ j=0,1,\cdots\}$. Let Σ_δ^ℓ be the δ-sampled system associated to Σ^ℓ. This is the discrete-time system associated to the pair $(e^{\delta A}, A^{(\delta)}B)$, again restricted to the above span. Call Σ^{ℓ} *δ-controllable* if Σ^{ℓ}_{δ} is controllable, which is in fact equivalent in the linear case to δ-accessibility at all points. We then have the following sufficient condition:

Lemma 17: Let Σ be a control-linear system with finite dimensional Lie algebra. Then, for each δ>0 for which Σ^{ℓ} is δ-controllable, $L_o = L_{\delta}$. In particular, if δ is like this and Σ satisfies the strong accessibility rank condition at ξ, it follows that Σ is δ-accessible at ξ.

There are many results in the literature regarding the characterization of δ-controllability for linear systems. See for instance [KHN] and [BL], [GH]. The simplest sufficient condition is that $\delta(\lambda-\mu) \neq 2k\pi i$ for all integers and all pairs of distinct eigenvalues of A, or even for pairs of eigenvalues of the restriction of A to the reachable set of Σ. (This eigenvalue condition, interestingly enough, appears also in Lie theory.) We now restrict attention to *bilinear* systems (see e.g. [MO].) These are described by equations $\dot{\xi} = (F + \Sigma u_i G_i)\xi$, where $M=\mathbb{R}^n$, $U=\mathcal{U}=\mathbb{R}^m$ and F and the G_i are nXn matrices. One is interested in accessibility at all $\xi \neq 0$. Viewing the above as a control-linear system with f and all g_i linear vector fields, the algebra L is a subalgebra of $gl(n,\mathbb{R})$, and A is the operator $C \longrightarrow CF-FC$, restricted to L. Thus, by matrix equation theory in linear systems, (or from a standard Lie algebraic fact, as in [JC], chapter I, exercise 16) the eigenvalues of A are differences of eigenvalues of F. Applying the linear result just quoted, we conclude that

Theorem 18: Let Σ be a bilinear system, and assume that δ>0 is such that

$\delta[(\lambda_1+\lambda_2)-(\mu_1+\mu_2)] \neq 2k\pi i$ for all nonzero $k \in \mathbf{Z}$

for any set of 4 eigenvalues $\lambda_1, \lambda_2, \mu_1, \mu_2$ of F. If Σ satisfies the strong accessibility rank condition at ξ then Σ is δ-accessible at ξ.

Note that, bilinear systems being analytic, δ-accessibility at ξ for *any* δ implies the rank condition. Roughly, the theorem says that sampling at twice the frequency predicted by the linear theory will insure preservation of accessibility. Actually, a much weaker condition is sufficient. By a simple coordinate change in U, we can replace F by any linear combination $F+\Sigma r_i G_i$. The eigenvalue condition for any of these new matrices will then be sufficient. Thus in various senses of genericity, all sampling periods δ preserve accessibility, for fixed F, if the G_i are "randomly" choosen. Finally, note that one could talk about systems like bilinear but with an added extra linear term on ξ ("internally biaffine systems"); an analogous eigenvalue condition results in that case as well.

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SOME RECENT RESULTS ON NONLINEAR FEEDBACK

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ABSTRACT

We proved earlier that stabilizable systems can be made input to state stable. In this paper we present a generalization of this fact to systems that are not necessarily linear in controls. For systems that are linear in controls, we also provide an explicit proof of Artstein's Theorem, and show how this results in an input-to-state stabilizer.

1 Introduction

In a previous paper ([10]) we studied the problem of when a system on $\mathbb{R}^n$,

$$\dot{x}(t) = f(x(t)) + u_1(t)g_1(x(t)) + \ldots + u_m(t)g_m(x(t)) \quad (1)$$

or in short-hand notation,

$$\dot{x} = f(x) + G(x)u \quad (2)$$

with f and the entries of the $n \times m$ matrix G being smooth (i.e., infinitely differentiable,) and $f(0) = 0$, can be made input to state stable –*ISS*– in a rather strong sense to be reviewed below. Our main result there was that this system is ***smoothly input to state stabilizable*** (that is, there exists a smooth map $k : \mathbb{R}^n \to \mathbb{R}^m$ with $k(0) = 0$ such that under the control law $u = k(x) + v$ the new system

$$\dot{x} = (f(x) + G(x)k(x)) + G(x)u$$

is ISS) ***if and only if*** the system (2) is ***smoothly stabilizable*** (that is, there exists an (in general different) k so that

$$\dot{x} = f(x) + G(x)k(x) \quad (3)$$

is globally asymptotically stable –*GAS*). The necessity is a trivial consequence of our definition of ISS, which implies GAS, but the converse is somewhat harder to establish. (By global asymptotic stability we mean the usual concept: attraction –solutions are defined for $t \geq 0$ and every initial state, and converge to 0– plus local asymptotic stability –initial states produce trajectories near the origin.)

One studies input to state stabilization because this is a more "robust" problem than stabilization. This type of design is important in dealing with questions of so-called coprime factorization for control systems (see later), as well as in dealing with possible noise in the implementation of control laws.

It is natural to ask if the same result can be proved for the more general system

$$\dot{x} = f(x, u) \quad (4)$$

which is not necessarily linear in u. More precisely, we assume that $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}^m$, that f is a differentiable (much less is needed) function from $\mathbb{R}^{n+m}$ into $\mathbb{R}^n$, and that 0 is an equilibrium point for the system, $f(0,0) = 0$.

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Unfortunately, the result does not generalize. As a counterexample, let $m = n = 1$ and consider the system

$$\dot{x} = -x + u^2x^2 \ .$$

We claim that for no possible feedback law k can there hold for

$$\dot{x} = -x + (k(x) + u)^2x^2 \quad (5)$$

that: (a) for $u \equiv 0$ the system is GAS, and (b) for $u \equiv 1$ and initial condition $x(0) = 4$ the solution remains bounded. Indeed, property (a) implies that $|k(x)| < 1/\sqrt{x}$ for all $x > 0$, but then this implies that the right-hand side of (5) is positive for all $x \geq 4$, so the solution with $x(0) = 4$ diverges to $+\infty$. So not even a very weak notion of input to state stability can be obtained for this example.

However, if one allows instead more general feedback laws, of the type

$$u = k(x) + G(x)v \quad (6)$$

where G is an $n \times n$ matrix of smooth functions invertible for all x, not necessarily the identity, then the theorem *can* be proved for the larger class of systems. This more general class of feedback law is of some interest, –for instance, it is the class used in most modern "geometric" nonlinear control,– although it is not appropriate for solving the type of problem (coprime factorizations) that was of interest in ([10]) if one insists on the definition given there for factorability. However, we shall also see below that with a somewhat weaker definition, as given for example in [7], coprime factorizations *can* be obtained using the feedback laws of type (6).

In this note we precisely state the above-mentioned general result, for (4) under feedback laws (6), and give as an application of this an alternative proof of a "folk" fact about dynamic extensions. We also provide a result which shows that GAS by itself is sufficient to guarantee a notion of ISS with "small controls," and give various applications of this result to local stabilization of cascaded systems and other problems. One of these corollaries shows, using only elementary techniques, that *any cascade* of locally asymptotically stable systems is again asymptotically stable; no assumptions need be made as is the case with center-manifold type of arguments. For details of the proofs see [11].

The second part of this paper deals with relaxing the regularity of feedback so as to allow for a possible lack of smoothness at the origin. The above results extend with no difficulty to this case, since the proofs do not involve differentiability at 0. However, the stabilization property becomes then much easier to deal with. In that part we then provide a simple, explicit, and in a sense "universal" proof of a result due to Artstein ([1]), and obtain certain generalizations of it.

The Artstein result concerns control systems of the type (1). It is assumed that there is given a *control-Lyapunov function* (henceforth just "*clf*") V for this system, that is, a smooth, proper, and positive definite function $V : \mathbb{R}^n \to \mathbb{R}$ so that

$$\inf_{u\in\mathbb{R}^m}\{L_fV(x) + u_1L_{g_1}V(x) + \ldots + u_mL_{g_m}V(x)\} < 0 \quad (7)$$

for each $x \neq 0$. In other words, V is so that for each nonzero state x one *can* diminish its value by applying *some* open-loop control. Recall that positive definite means that $V(0) = 0$ and $V(x) > 0$ for $x \neq 0$, and proper means that $V(x) \to \infty$ as $\|x\| \to \infty$.

It is easy to show that the existence of such a clf implies that the system is asymptotically controllable (from any state one can asymptotically reach the origin); in the paper [8] we showed that that the existence of a clf is in fact also *necessary* if there is asymptotic controllability, provided that one does not require smoothness (in which case equation (7) must be replaced by an equation involving Dini derivatives). More relevant to the topic of this paper, it was shown in [1] that if there is a clf, smooth as above, then there must be a feedback law $u = k(x)$, $k(0) = 0$ which globally stabilizes the system and which is smooth on $\mathbb{R}^n_0 := \mathbb{R}^n - 0$. In general k may fail to be smooth everywhere, but under certain conditions, which we study below, k can be guaranteed to be at least continuous at the origin in addition to being smooth everywhere else. (The conditions are necessary as well as sufficient, the necessity following from the by now standard Lyapunov function inverse theorems due to Massera, Zubov, Kurzweil and others.) In that case, the material in the first half of the paper (input to state stabilization) applies.

Allowing nonsmoothness at the origin was emphasized in [14], and has since been recognized as desirable by many authors (see e.g. [15], [5], and [2], the first of which in particular also proved a version of Artstein's theorem). From the point of view of Lyapunov techniques, this is more natural than global smoothness, because it can be characterized precisely in terms of Lyapunov functions.

Since (control-) Lyapunov functions are as a general rule easier to obtain than the feedback laws themselves –after all, in order to prove that a given feedback law stabilizes, one often has in addition to provide a suitable Lyapunov function anyway,– Artstein's theorem provides in principle a very powerful approach to nonlinear stabilization. Previous proofs relied on nonconstructive partition of unity arguments. To make it a practical technique, one needs a more explicit construction. In this note we give one such explicit, very elementary, construction of k from V and the vector fields defining the system. A further advantage of our method, in addition to its extreme simplicy and ease of implementation, is that it provides automatically an *analytic* feedback law if the original vector fields as well as the clf are also analytic. In [12] we prove also a (less elementary, but in some senses weaker) theorem that shows that if V as well as $f, g_1, \ldots, g_m$ are *rational* then k can be chosen also rational.

Our construction is based on the following observation, which for introductory purposes we restrict to single-input ($m = 1$) systems only. Assume that V is a clf for the system $\dot{x} = f(x) + ug(x)$. Denote $a(x) := \nabla V(x).f(x)$, $b(x) := \nabla V(x).g(x)$. The condition that V is a clf is precisely the statement that

$$b(x) \neq 0 \Rightarrow a(x) < 0$$

for all nonzero x. In other words, for each such x,

the pair $(a(x), b(x))$ is stabilizable

when seen as a single-input, one-dimensional linear system. On the other hand, giving a feedback law $u = k(x)$ for the original system, with the property that the same V is a Lypaunov function for the obtained closed-loop system

$$\dot{x} = f(x) + k(x)g(x)$$

is equivalent to asking that

$$\nabla V(x).(f(x) + k(x)g(x)) < 0 \ ,$$

that is,

$$a(x) + k(x)b(x) < 0 \ ,$$

for all nonzero x. In other words, $k(x)$, seen as a 1×1 matrix, must be a constant linear feedback stabilizer for $(a(x), b(x))$, for each fixed x.

We now interpret $(a(x), b(x))$ as a *family of linear systems parameterized by x*. This family depends smoothly on x. From the theory of families of systems or "systems over rings" we know that since each such linear system is stabilizable there exists indeed a smoothly dependent k as wanted. Moreover, this k can be chosen to be analytically or rationally dependent if the original family is, that is, if the original system and clf are. The general theory is surveyed in [9], and the result in the smooth and analytic cases is due to Delchamps (see for instance [3]), but in this very simple case (the family is one-dimensional), the construction of k can be carried out directly without explicit recourse to the general result. Indeed, one can show directly that the following feedback law:

$$k := -\frac{a + \sqrt{a^2 + b^2}}{b}$$

works. This results from the solution of an LQ problem, and is analytic, in fact algebraic, on a, b. (The apparent singularity due to division by b is removable, as discussed later.) Along trajectories of the corresponding closed-loop system, one has that

$$\frac{dV}{dt} = -\sqrt{a^2 + b^2} < 0$$

as desired. This feedback law may fail to be continuous at zero, however. If one modifies it slightly to

$$\boxed{k := -\frac{a + \sqrt{a^2 + b^4}}{b}}$$

then under the natural hypotheses –reviewed later– this does become continuous. Except for the rational case which must be treated separately, and for the multi-input case which is an immediate generalization, this will be the final form of our feedback law which provides a proof of Artstein's theorem.

2 Results on Input to State Stability

We recall the basic terminology from ([10]). The function $\gamma : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is *of class $\mathcal{K}$* if it is continuous strictly increasing and satisfies $\gamma(0) = 0$; it is *of class $\mathcal{K}_\infty$* if in addition $\gamma(s) \to \infty$ as $s \to \infty$. If γ is of class $\mathcal{K}_\infty$ then the inverse function γ^{-1} is well defined and is again of class $\mathcal{K}_\infty$. A function $\beta : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is *of class $\mathcal{KL}$* if for each fixed t the mapping $\beta(\cdot, t)$ is of class $\mathcal{K}$ and for each fixed s it is decreasing to zero on t as $t \to \infty$. We use single bars $|\xi|$ to denote Euclidean norm of states and controls, and use $\|u\| := \text{ess.sup.}\{|u(t)|, t \geq 0\}$ for measurable essentially bounded controls.

The system (4) is *globally asymptotically stable (GAS)* if there exists a function $\beta(s, t)$ of class $\mathcal{KL}$ such that, with the control

$u \equiv 0$, given any initial state ξ_0 the solution exists for all $t \geq 0$ and it satisfies the estimate

$$|x(t)| \leq \beta(|\xi_0|, t).$$

The system is *input to state stable (ISS)* if there is a function β of class $\mathcal{KL}$ and there exists a function γ of class $\mathcal{K}$ such that for each measurable essentially bounded control $u(\cdot)$ and each initial state ξ_0, the solution exists for each $t \geq 0$ and furthermore it satisfies

$$|x(t)| \leq \beta(|\xi_0|, t) + \gamma(\|u\|). \tag{8}$$

Since $\gamma(0) = 0$, an ISS system is necessarily GAS; the later is equivalent to the usual notion of asymptotic stability ("$\varepsilon - \delta$" stability plus attractivity). The former says basically that for bounded initial state and control, a bounded trajectory results, and further (since β decays) that eventually the state is bounded by a function of the control alone (and this bound is small if the control is small). This is much stronger that asking GAS plus "bounded-input bounded-state" stability. To see this, take the system

$$\dot{x} = (\sin^2 u - 1)x \ .$$

This is "BIBS", and with control $u \equiv 0$ one has $\dot{x} = -x$ which is globally stable. However, with $u \equiv \pi/2$ trajectories do *not* become ultimately bounded by a constant independent of ξ_0. Since we prove mainly positive results, chosing such a strong notion of stability makes the results more informative. (The one negative example given earlier, dealing with restrictive types of feedback laws, was such that not even the weakest possible version of these properties holds.)

We shall say that the system (4) is *weakly input to state stabilizable* if there exists a continuously differentiable map $k : \mathbb{R}^n \to \mathbb{R}^m$ with $k(0) = 0$ and an $n \times n$ matrix G of continuously differentiable functions, invertible for each x, such that under the control law (6) (and replacing v by u,) the system

$$\dot{x} = f(x, k(x) + G(x)u) \tag{9}$$

is ISS. We say that the system is *stabilizable* if there is a differentiable k such that this new system (with $G \equiv 0$) is GAS. The first property implies the second. The next result shows they are equivalent. The terminology "weak" is used to avoid confusion with the notion in ([10]), where G = identity is required.

Theorem 1 *For* (4) *under control laws of the type* (6), *stabilizability implies weak input to state stabilizability.*

To prove this theorem we may assume that the original system is already GAS, and we look for a G so that $\dot{x} = f(x, G(x)u)$ is ISS. The idea of the proof can be easily understood in the case of one-dimensional systems ($n = 1$): Since the system is GAS, it is necessary in that case that

$$xf(x, 0) < 0$$

for all nonzero x. Thus by continuity also $xf(x, u) < 0$ for small u. So it is only necessary to multiply controls by a very small number in order to have states approach the origin. The same idea works in higher dimensions, using Lyapunov functions, but working out the details takes some effort. The proof is given in [11]; it is far less constructive than the result given in [10].

The one dimensional case also suggests that as long as the controls remain small, ISS may hold. This idea gives rise to our other result:

Theorem 2 *If the system* (4) *is GAS then there exist a function β of class $\mathcal{KL}$, a function γ of class $\mathcal{K}$, and a continuous function $\sigma : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$, $\sigma(s) \neq 0$ if $s \neq 0$, such that, for each initial state ξ_0 and for each measurable essentially bounded control $u(\cdot)$ for which*

$$\|u\| \leq \sigma(|\xi_0|)$$

the solution of (4) *exists for each $t \geq 0$ and it satisfies* (8).

3 Further Remarks

In [16] (see also e.g. [6]) it is proved that under certain hypotheses GAS implies ISS (or more precisely, a notion of bounded-input bounded-state stability). The hypotheses are basically that the map f be globally Lipschitz on x, uniformly in u, (if the system has the form $\dot{x} = f(x) + G(x)u$ this is insured provided that the control term $G(x)$ is bounded,) and that stability be *exponential*. Our first remark is that no such result holds if one has only non-exponential stability. To illustrate this point it is sufficient to consider the system

$$\dot{x} = -\tanh x + u \ .$$

This is GAS, and $f(x, u)$ is globally Lipschitz, but with $u \equiv 1$ one has that all trajectories diverge to $+\infty$.

It was remarked in [10] that a cascade composition of what are there called "input/output stable" maps is again of that kind. In terms of state-space notions only, this translates into the fact that cascading a GAS system with an ISS one gives a GAS system. More precisely, consider the composite system

$$\dot{x} = f(x, z) \tag{10}$$
$$\dot{z} = g(z) \tag{11}$$

where the second is GAS and the first is ISS (with z as control). The claim is that the composite system is GAS. Indeed, if β_1 and γ are as in the definition of ISS for the first system, and β_2 is as in the definition of GAS for the second, then for each trajectory necessarily

$$|x(t)| + |z(t)| \leq \beta(|x(0)| + |z(0)|, t)$$

where

$$\beta(s, t) := \beta_2(s, t) + \beta_1(\beta_1(s, \frac{t}{2}) + \gamma(\beta_2(s, \frac{t}{2})), \frac{t}{2}) + \gamma(\beta_2(s, \frac{t}{2})))$$

is again of class $\mathcal{KL}$.

If the first system is just GAS rather than ISS, there is no reason for the cascade to be again GAS, except locally (see below). However, the proof of theorem 1 gives some $G(x)$ so that $\dot{x} = f(x, G(x)u)$ is ISS. Changing coordinates

$$(x, z) \mapsto (x, y) \ , \ \ y := G(x)^{-1}z$$

there results a new composite system

$$\dot{x} = f(x, G(x)y)$$
$$\dot{y} = \tilde{g}(x, y)$$

where the first system is now ISS. If it were possible to modify $\tilde{g}$ in some manner so that it becomes independent of x and the second system is GAS, then the above argument gives that the composite system (even in the original coordinates) is GAS. This program can be carried out in the following situation: Assume that z has dimension 1 and one considers the cascade

$$\dot{x} = f(x, z)$$
$$\dot{z} = u$$

(a "dynamic extension" of the first system). Then the above change of coordinates results in a second equation of the type

$$\dot{z} = h(x,z)u + k(x,z)$$

where h is always nonzero. Applying the feedback law

$$u = \frac{1}{h(x,z)}(-k(x,z) - z)$$

results then in a composite GAS system. We thus recover the folk result (see e.g. [13] for an application and references) that if a system is smoothly stabilizable then adding an integrator does not change this property. ∎

Finally, we give an application of theorem 2 to show that, for each compact subset of the state space, small enough controls tending to zero give rise to trajectories that also converge to zero. This is a local version of the corresponding global result given in [10], with basically the same proof.

Corollary 3.1 With the notations in theorem 2, for each real number $k > 0$ the following property holds. For each control u such that $||u|| < \sigma(k)$ and each initial $|x_0| \leq k$, the solution of (4) exists for all $t > 0$ and it satisfies $\lim_{t\to 0} x(t) = 0$.

Proof. Let $l := \beta(k,0) + \gamma(\sigma(k)) \leq k$, and pick any u and $x(\cdot)$ as above. We claim that $|x(t)| \leq l$ for all t (and in particular then, the trajectory is defined globally). Indeed, if the trajectory ever exits the ball of radius k then there is a first T so that $|x(T)| = k$. Consider in that case the restriction of u to times $t \geq T$,

$$v := u|_{[T,+\infty)} \tag{12}$$

which has norm again $< \sigma(k)$. Then, by theorem 2,

$$|x(t)| \leq \beta(|x(T)|, t - T) + \gamma(||v||) \tag{13}$$

which is less than l for all t.

Pick any $\varepsilon > 0$; we wish to show that $|x(t)| < \varepsilon$ for all large t. Let $c > 0$ be the minimum of σ on the interval $[\varepsilon, l]$. Since $u(t) \to 0$, there is some t_0 such that $\gamma(||v||) < \varepsilon$ and also $||v|| < c$ for all restrictions v as in (12) and all $T \geq t_0$. If it happens that $|x(t)| < \varepsilon$ for all $t \geq t_0$ then there is nothing left to prove. Otherwise, there is some $T \geq t_0$ so that $\varepsilon \leq |x(T)| \leq l$. For this T, equation (13) holds. Since the first term tends to zero and the second is less than ε, it follows that $|x(t)| < \varepsilon$ for large t, as desired. ∎

Together with the argument used in proving the stability of the cascade composition (10)-(11), it follows then that, if (10) is GAS when $z \equiv 0$ and (11) is locally asymptotically stable *(LAS)*, then the composed system (10)-(11) is also LAS. Indeed, we may fix any k, and then pick a $0 < \delta < \sigma(k)$ for which all trajectories of (11) starting at every $|z_0| < \delta$ necessarily satisfy $z(t) \to 0$ and $|z(t)| < \sigma(k)$. Then the corollary guarantees that x also goes to zero, and moreover that it remains bounded by $l = l(k)$, which is small for small k. In fact, (we owe this observation to Héctor Sussmann), it is enough to suppose that the first system (10) is itself *locally* stable because one can just restrict the system to the domain of attraction of the origin (which is diffeomorphioc to Euclidean space). We conclude the following fact, proved using very elementary techniques:

Corollary 3.2 If (10) is LAS when $z \equiv 0$, and if (11) is also LAS, then the composite system (10)-(11) also is. □

4 Coprime Factorizations

We refer the reader to [10] for the definitions of input/output stable (IOS) operators, and the basic properties of this concept. We now modify Definition 4.1 (more precisely, its equivalent formulation via Lemma 4.2) in the above reference, to weaken the concept of coprimeness:

Definition 4.1 The i/o operator $P : \mathcal{D}(P) \to L^p_{\infty,e}$ *admits a weakly coprime right factorization* if there exist IOS operators $A : L^m_{\infty,e} \times L^p_{\infty,e} \to L^m_{\infty,e}$, $N : L^m_{\infty,e} \to L^p_{\infty,e}$, and $D : L^m_{\infty,e} \to L^m_{\infty,e}$, such that D is causally invertible, $\mathcal{D}(D^{-1}) = \mathcal{D}(P)$, $P = ND^{-1}$ and, if I denotes the identity in $L^m_{\infty,e}$,

$$A \circ \begin{pmatrix} D \\ N \end{pmatrix} = I\,.$$

Theorem 3 *If (4) is smoothly stabilizable then its input to state mapping admits a weak coprime right factorization.*

Proof. We first use Theorem 1 to obtain G and k. Now, let P be the operator defined as follows:

$$P\begin{pmatrix} v(\cdot) \\ x(\cdot) \end{pmatrix}(t) \;:=\; G^{-1}(x(t))[-k(x(t)) + v(t)]\;.$$

Since it is memoryless, and G and k are continuous everywhere, it follows that P is IOS. The operators N and D are chosen precisely as in [10], that is, N is the input to state mapping of the closed-loop system (9), and D is the resulting mapping $u \mapsto k(x) + G(x)u$, which is itself stable (for the same reason that P is) and invertible. ∎

5 Artstein's Theorem

We start with some definitions for the system (2), which relax smoothness at the origin.

Definition 5.1 Let $k : \mathbb{R}^n \to \mathbb{R}^m$ be a mapping, smooth on $\mathbb{R}^n_0$ and with $k(0) = 0$. This is a *smooth feedback stabilizer* for the system (2) if (3) is GAS. □

The fact that k may fail to be even continuous at the origin causes no problems regarding uniqueness of solutions, as is easy to verify from the definition of asymptotic stability. A sufficient (as well as necessary) condition for a given k to be a smooth feedback stabilizer is that there exist a Lyapunov function for the closed-loop system, i.e. a smooth, proper, and positive definite function V so that

$$\nabla V(x).[f(x) + k_1(x)g_1(x) + \ldots + k_m(x)g_m(x)] \;<\; 0$$

for all nonzero x. Observe that such a Lyapunov function is automatically a clf for the open-loop system (1). Note also that if k happens to be continuous at the origin then the following property holds too (with $u := k(x)$):

For each $\varepsilon > 0$ there is a $\delta > 0$ such that, if $x \neq 0$ satisfies $||x|| < \delta$, then there is some u with $||u|| < \varepsilon$ such that

$$\nabla V(x).f(x) + u_1 \nabla V(x).g_1(x) + \ldots + u_m \nabla V(x).g_m(x) \;<\; 0\;.$$

We shall call this the *small control property* for the clf V. The existence of a clf with this property is necessary if there is any smooth stabilizer continuous at zero; part of Artstein's theorem is the statement that this is also sufficient. Thus we wish to prove the following result:

Theorem 4 *If there is a smooth clf V (respectively, the system as well as V are real-analytic,) then there is a smooth (respectively, real-analytic) feedback stabilizer k. If V satisfies the small control property, then k can be chosen to be also continuous at* 0.

Proof. We shall prove the theorem by constructing, once and for all, a fixed real-analytic function φ of two variables, and then designing the feedback law in closed-form, analytically, from the evaluation of this function at a point determined by $\nabla V(x).f(x)$ and the $\nabla V(x).g_i(x)$'s.

Consider the following open subset of $\mathbb{R}^2$, which can be interpreted as a subset of the set of all stabilizable single-input linear systems of dimension one:

$$S := \{(a,b) \in \mathbb{R}^2 \mid b > 0 \text{ or } a < 0\}$$

Pick any real analytic function $q : \mathbb{R} \to \mathbb{R}$ such that $q(0) = 0$ and $bq(b) > 0$ whenever $b \neq 0$. (Later we specialize to the particular case $q(b) = b$.) We now show that the function defined by

$$\varphi(a,0) := 0 \text{ for all } a < 0$$

and

$$\varphi(a,b) := \frac{a + \sqrt{a^2 + bq(b)}}{b}$$

otherwise, is real-analytic on S. For this, consider the algebraic equation

$$F(a,b,p) = bp^2 - 2ap - q = 0 \qquad (14)$$

which is satisfied by $p = \varphi(a,b)$ for each $(a,b) \in S$. We show that the derivative of F with respect to p is nonzero at each point of the form $(a,b,\varphi(a,b))$ with $(a,b) \in S$, from which it will follow by the implicit function theorem that φ must indeed be real-analytic. Indeed,

$$(1/2)\frac{\partial F}{\partial p} = bp - a$$

equals $-a \neq 0$ when $b = 0$ and $\sqrt{a^2 + bq(b)} \neq 0$ otherwise.

Assume that V is a clf. As in the introduction, we let

$$a(x) := \nabla V(x).f(x), \quad b_i(x) := \nabla V(x).g_i(x), \ i = 1,\ldots,m \ .$$

We also let

$$B(x) := (b_1(x),\ldots,b_m(x)), \ \ \beta(x) := \|B(x)\|^2 = \sum_{i=1}^{m} b_i^2(x) \ .$$

Then the condition that V is a clf is again equivalent to asking that $\beta(x) = 0$ imply $a(x) < 0$, that is, that $(a(x),\beta(x)) \in S$ for each nonzero x, or equivalently, that the one-dimensional time-invariant systems with m controls $(a(x),B(x))$ be stabilizable, for each such x. Thus we may define the feedback law $k = (k_1,\ldots,k_m)$, where:

$$k_i(x) \ := \ -b_i(x)\,\varphi(a(x),\beta(x))$$

for $x \neq 0$ and $k(0) := 0$. This is smooth, and it is also real-analytic if V as well as $f, g_1, \ldots, g_m$ are. Moreover, at nonzero x we have that

$$\begin{aligned}\nabla V(x).\left[f(x) + k_1(x)g_1(x) + \ldots + k_m(x)g_m(x)\right]& \\ &= a(x) - \varphi(a(x),\beta(x))\beta(x) \\ &= -\sqrt{a(x)^2 + \beta(x)q(\beta(x))} \ < \ 0\end{aligned}$$

so the original V decreases along trajectories of the corresponding closed-loop system, and is a Lyapunov function for this.

Finally, assume that V satisfies the small control property. We wish to show that the function k is continuous at the origin. Pick any $\varepsilon > 0$. We will find a $\delta > 0$ so that $\|k(x)\| < \varepsilon$ whenever $\|x\| < \delta$. Since $k(x) = 0$ whenever $\beta(x) = 0$, we may assume that $\beta(x) \neq 0$ in what follows. We also take $q(b) := b$ from now on, for simplicity. Let $\varepsilon' := \varepsilon/3$.

Since V is positive definite, it has a minimum at 0, so $\nabla V(0) = 0$. Since the gradient is continuous, it holds that each of the $b_i(x)$ are small when x is small. Together with the small control property, this means that there is some $\delta > 0$ such that, if $x \neq 0$ satisfies $\|x\| < \delta$, then there is some u with $\|u\| < \varepsilon'$ so that $a(x) + B(x)u < 0$ and in addition $\|B(x)\| < \varepsilon'$. The first of the above implies, by the Cauchy-Schwartz inequality, that $a(x) < \varepsilon'\|B(x)\|$ if $0 < \|x\| < \delta$, and so also $|a(x)| < \varepsilon'\|B(x)\|$ in the case when $a(x) > 0$. On the other hand, observe that

$$b\varphi(a,b) = a + \sqrt{a^2 + b^2} \le 2|a| + b$$

for all $(a,b) \in S$ with $b > 0$. Thus, if $0 < \|x\| < \delta$ and $a(x) > 0$, necessarily

$$\varphi(a(x),\beta(x)) \le \frac{2\varepsilon'}{\|B(x)\|} + 1$$

and hence also

$$\|k(x)\| \ = \ \varphi(a(x),\beta(x))\,\|B(x)\| \ \le \ 3\varepsilon' = \varepsilon$$

as desired. There remains the case of those x for which $a(x) \le 0$. There, $0 < a(x) + \sqrt{a(x)^2 + \beta(x)^2} \le \beta(x)$ so $0 \le \varphi(a(x),\beta(x)) \le 1$ and therefore

$$\|k(x)\| = \varphi(a(x),\beta(x))\|B(x)\| \le \varepsilon' < \varepsilon$$

as desired too. ∎

Next we remark how the form of the function φ could be obtained from an optimization problem. For each $x \neq 0$ one can think of the linear system (writing $b_i := b_i(x)$ and $a := a(x)$):

$$\dot{z} = az + \sum_{i=1}^{m} b_i u_i = az + Bu$$

and this is stabilizable in the usual linear sense. One needs a stabilizing feedback law k, thought of now as a constant row vector (for each fixed x), that is a k so that $a + \sum_{i=1}^{m} b_i k_i < 0$. If we pose the linear-quadratic problem of minimizing

$$\int_0^\infty u^2(t) + qz^2(t)\,dt$$

the solution is given by the feedback law $u := -B'pz$, where p is the positive solution of the algebraic Riccati equation (14). Thus we recover $k_i = -b_i\varphi(a,b)$. The fact that this is analytic on the data is no accident; Delchamp's theorem ([3]) guarantees this even for systems in arbitrary dimensions, essentially by the same argument we gave for scalar equations. The choice $q := b$ is made so that, in the above cost, the u^2 term is given more relative weight when b is small, forcing small controls when b is small.

Note that the form of our feedback law,

$$k(x) = -\frac{a + \sqrt{a^2 + \beta^2}}{\beta}B'$$

is closely related to the feedback

$$k(x) = -\frac{a + \beta}{\beta}B'$$

proposed in [15] under the particular assumption that β never vanishes, as well as the law $k(x) = -B'$ used in [4] (and related papers) when a can be guaranteed to be nonpositive and a certain transversality condition holds.

In connection with the material in the previous sections, the feedback "correction" needed after stabilization was given in [10] by the simple formula

$$k(x) := -\frac{\tilde{a}(x)}{2m} B(x)'$$

where $\tilde{a}$ is $\nabla V.\tilde{f}$ for the closed-loop $\tilde{f}$ that resulted from first stabilizing the system in the usual sense. Together with the results in this paper, the following formula is obtained for the input-to-state stabilizer k if a clf V exists:

$$k := -\left(\frac{a + \sqrt{a^2 + \beta^2}}{\beta} + \frac{\sqrt{a^2 + \beta^2}}{2m} \right) B'$$

(dropping x's). To be more accurate, a mild technical condition is needed for the result in [10], which in this case becomes the assumption that

$$\lim_{x \to \infty} a^2 + \beta^2 = \infty \ .$$

If this would not hold, that reference shows how to slightly modify the construction.

Finally, we note that in the case in which V is merely a "weak" clf, meaning that $\beta = 0$ implies just that $a \leq 0$ (not necessarily $a < 0$), the LaSalle invariance principle could be used to guarantee stabilization, provided that there are no nonzero points where β and f vanish simultaneously.

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ALGEBRAIC GEOMETRIC METHODS IN REALIZATION OF DISCRETE-TIME NONLINEAR SYSTEMS

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ABSTRACT
This paper studies the realization theory of Volterra series by polynomial and related classes of systems. Existence and uniqueness of canonical realizations is proved, and finiteness results are described.

(1) INTRODUCTION

We consider in this paper the realization theory of nonlinear input/output maps defined by (discrete-time) Volterra series. These are the polynomial input/output maps, in which sequences of m-dimensional inputs $u(j)$ are mapped into sequences of p-dimensional outputs

$$y(t) = \sum_{i=0}^{\infty} L_{it}(u(0),\ldots,u(t-1))$$

for all $t \geq 0$, where each term L_{it} is a degree i homogeneous polynomial in mt variables and, for each t, only finitely many $L_{it} \neq 0$. (The L_{it} will be subject to a mild and natural technical restriction, described below.) We are in principle interested in realizations by polynomial systems, (where next-state and output maps are multivariable polynomial functions,) but a somewhat more general class will have to be defined in order to develop a satisfactory theory.

The main result for i/o maps as above is the existence and uniqueness of "canonical" realizations. Characterizations are given for the cases in which canonical realizations are finite-dimensional and/or polynomial systems.

The results here presented are a generalization of the equilibrium-initial state case discussed in SONTAG [1976 a]; in most cases we only sketch proofs, since these are parallel to the latter. Here we have taken a category-theoretical approach, which seems most natural here. (The general categorical realization methods, e.g. ARBIB and MANES [1974] are not applicable, due to the nature of the categories appearing in this context.) Since doing so doesn't increase the difficulty, we shall consider coefficients belonging to an arbitrary field k (not necessarily reals or complexes), to be fixed for the rest of this discussion.

(2) RESPONSE MAPS

Giving a polynomial input/output map is equivalent to specifying a polynomial response map $f\colon U^* \to Y$, where $U = k^m$ and $Y = k^p$, f mapping a sequence $u(0),\ldots,u(t-1)$ into the corresponding output at time t. This map f is not an algebraic object, so we shall first show how to extend f to a map $f\colon \Omega \to Y$, for a suitable "completion" of U^*, Ω a generalized type of algebraic variety (an affine scheme) and f now a "polynomial map" (scheme morphism). Before proceeding, we make the assumption that the degree of $y(t)$ in $u(t-j)$ depends only on j. This is not the assumption resulting in the restriction to "bounded maps" in SONTAG [1976 b] and (5) below: that case requires a uniform bound independent of j. The present restriction is a very weak one which will obviously hold whenever one considers i/o maps of polynomial systems. On the other hand, we shall prove our results for U,Y more general than the above (in particular any algebraic varieties) so that polynomial constraints in input and output values sets may be implicitly included in system descriptions.

(3) REDUCED OBJECTS

We shall fix an algebraic category $\underline{\underline{V}}$; thus $\underline{\underline{V}}$ is an equational variety of universal algebras. The example to bear in mind is $\underline{\underline{V}}$ = all k-algebras over the field k. We assume the initial object I to be nonempty, i.e. that there exist constant operations (in the above example, $I=k$). Morphisms $e\colon C \to I$, C in $\underline{\underline{V}}$, will be called evaluations. An evaluation induces, for each A in $\underline{\underline{V}}$, a morphism e_A defined by the diagram

$$\begin{tikzcd} C \arrow[r, "e"] \arrow[d, "i_2"'] & I \arrow[d] \\ A \amalg C \arrow[r, "e_A"] & A \end{tikzcd},$$

where $\amalg$ indicates coproduct. In the k-algebra example, coproduct is tensor product and the e_A are evaluations of polynomials with coefficients in A and "variables" in C. If there is at least one $e\colon C \to I$, the identity of A factors as $A \cong A \amalg I \to A \amalg C \to A$, the last morphism being e_A. Thus A is naturally included in $A \amalg C$. Similarly, the coproduct preserves monics in such a situation.

(3.1) DEFINITION: C in $\underline{\underline{V}}$ is a reduced object if: (a) for all A in $\underline{\underline{V}}$ and all x,y in $A \amalg C$, if $e_A(x) = e_A(y)$ for all $e\colon C \to I$, then $x = y$, and (b) if B is a subalgebra of A in $\underline{\underline{V}}$ and if x in $A \amalg C$ is such that $e_A(x)$ belongs to B for all e, then x is in $B \amalg C$.

When $\underline{\underline{V}}$ = all k-algebras and C is the coordinate ring of an algebraic variety, or more generally a k-reduced algebra, C is a reduced object (SONTAG [1976 a, "Main lemma" (10.7)]). The following properties, proof of which are standard, will be needed later:

(3.2) PROPOSITION: Let C be a reduced object. Then: (a) for any family $\{A_i\}$ of algebras in $\underline{V}$, the canonical map $(\Pi A_i) \amalg C \to \Pi(A_i \amalg C)$ is one-to-one; (b) if the A_i are all subalgebras of a fixed A, then, as subalgebras of $A \amalg C$, $(\bigcap A_i) \amalg C = \bigcap (A_i \amalg C)$, and (c) C_n, the coproduct n times of C by itself, is also reduced, for all $n \geq 0$.

We shall be interested in system realization in the dual category $\underline{V}^o$, since the latter is equivalent to the category of affine k-schemes in the example $\underline{V}$ = k-algebras. Motivated by the above example, we denote the contravariant duality functors by $A: \underline{V}^o \to V$ and Spec: $\underline{V} \to \underline{V}^o$. A reduced object C will be fixed for the rest of this paper; we denote U := Spec C = input value space. Using the separation properties in (3.1), there is a functor from the full subcategory of duals of reduced objects into Sets which permits identifying U with the set of evaluations $\underline{V}(C,I)$. In particular, there is a well-defined object U^* = free monoid on$_o$ U. Also fixed will be an arbitrary Y in $\underline{V}^o$, the output-value space.

(4) RESPONSES AND SYSTEMS:

We denote C_* := product of all the C_n, $n \geq 0$. The canonical morphism in (3.2.a) induces a morphism

$$\alpha: C_* \amalg C \to \prod_{n \geq 0} (C_n \amalg C) = \prod_{n \geq 1} C_n.$$

We define inductively the subalgebras of C_*: $C_{(0)} := C_*$, $C_{(n)} := (1 \times \alpha)(I \times C_{(n-1)} \amalg C)$, and C_∞ the intersection of all $C_{(n)}$. It follows from (3.2.b) that one has a fixed point $C_\infty = (1 \times \alpha)(I \times (C_\infty \amalg C))$. Thus

$\beta: pr_2 \circ (1 \times \alpha)^{-1}: C_\infty \to C_\infty \amalg C$ is well-defined, and δ := Spec β endows Ω := Spec C_∞, the input space, with a concatenation product. Dualizing the $C_\infty \to C_n$ obtained from the projections $C_* \to C_n$, we conclude that there are canonical maps $U^n \to \Omega$ and $U^* \to \Omega$. Furthermore, U^* is "dense" in Ω, in that morphisms from Ω are determined by their restrictions to U^*.

(4.1) DEFINITION: A response is a morphism $f: \Omega \to Y$.

This definition is justified by the fact that a polynomial response map $U^* \to Y$ as in (2) extends uniquely to a morphism as in (4.1), and, conversely, the restriction to U^* of a response is a polynomial response map when $U = k^m$, $Y = k^p$. The following provide a generalization of polynomial systems, the latter corresponding to the case $\underline{V}$ = k-algebras, $A(X)$ finitely generated:

(4.2) DEFINITION: A system $\Sigma = (X,P,x_o,h)$ is given by an X in $\underline{V}^o$ (the state-space), and morphisms $P: X \times U \to X$ (transitions), x_o: Spec $I \to X$ (initial state) and $h: X \to Y$ (read-out).

The n-th iterate $P^{(n)}: X \times U^n \to X$ is defined inductively ($P^{(o)}$ = identity), and the n-step reachability map $g_n: U^n \to X$ is the composition, using $P^{(n)}$, $U^n \simeq (\text{Spec } I) \times U^n \to X \times U^n \to X$. It follows from the constructions that $\{g_n\}: U^* \to X$ extends to a morphism $g: \Omega \to X$, the reachibility morphism of Σ. Thus to a system Σ there corresponds a well-defined response map realized by Σ, $f_\Sigma := h \circ g$. The realization problem is that of studying realizations of a given response map.

A system is quasi-reachable whenever $A(g)$ is one-to-one; this corresponds in the k-algebra example to g having a dense image in the Zariski topology. Given any Σ, restriction to the "closure of the reachable set" gives a quasi-reachable subsystem Σ_Q; more rigorously, X_Q is Spec $(A(X)/\ker A(g))$, where $\ker A(g)$ is the kernel congruence of $A(g)$, and P_Q, $(x_o)_Q$, h_Q are the naturally defined morphisms.

We introduce now the concept of observability. The translation by a w in U^* is the morphism $L_w: \Omega \to \Omega$ obtained by dualizing

$$(4.3) \quad C_\infty \to C_\infty \amalg C_n \to C_\infty \amalg I \simeq C_\infty,$$

where the first morphism is the n-th iterate of β and the second one is $1 \amalg w$. The corresponding observable h^w of a system Σ is the dual of $(1 \amalg w) \circ A(P^{(n)}) \circ A(h): A(Y) \to A(X)$; the observability map of Σ is $H: X \to \Gamma$, (where Γ is the product of Y, one copy for each w in U^*), induced by the observables h^w. The system is algebraically observable whenever $A(H)$ is onto. In the k-algebra case, this corresponds to each state coordinate being computable from the input/output data using only algebraic operations, see SONTAG and ROUCHLEAU [1975]. We can also define Σ to be abstractly observable whenever $A(H)$ is epi in the full subcategory of reduced objects; this corresponds to the standard notion of pairwise state distinguishability. If Σ is not algebraically observable, we can obtain a "quotient" system Σ^{obs} by defining $X^{obs} := \mathrm{Spec}\ (A(H)(\Gamma))$ and utilizing the fact (from 3.1.b) that $A^{obs} := A(H)(\Gamma)$ satisfies $A(P)A^{obs} \subseteq A^{obs} \amalg C$. (This is the point where C = reduced is crucial.)

A system will be canonical when quasi-reachable and algebraically observable.

(5) CANONICAL REALIZATIONS

Systems realizing a given response form a category using as morphisms $T: \Sigma_1 \to \Sigma_2$ the $T: X_1 \to X_2$ for which $T \circ x_0^{(1)} = x_0^{(2)}$, $T(P(\cdot,u)) = P(T(\cdot),u)$ for all u, and $h_2 \circ T = h_1$. (Note that existence of such a T in fact forces equality of responses.) In particular, there is for each Σ a decomposition diagram

(5.1)
$$\begin{array}{ccc} \Sigma_Q & \longrightarrow & \Sigma \\ \downarrow & & \downarrow \\ (\Sigma_Q)^{obs} = (\Sigma^{obs})_Q & \longrightarrow & \Sigma^{obs} \end{array}$$

where the horizontal morphisms are duals of onto and the rows are duals of one-to-one maps. Thus canonical realizations $\Sigma_f = (\Sigma_Q)^{obs}$ of f will exist as soon as a single realization Σ is shown to exist. The latter can be taken to be the free realization $\Sigma_{free}(f)$ with $X := \Omega$, $P := \delta$, $h := f$ and initial state the dual of the 0-th projection $C_\infty \to C_0 = I$. As in the linear and in the automata case (KALMAN, FALB and ARBIB [1969]), one has the

(5.2) THEOREM: Any response f has a canonical realization Σ_f, unique up to isomorphism.

PROOF: Existence is clear from (5.1) applied to $\Sigma_{free}(f)$. To prove uniqueness, it is enough to notice that the set of responses is in bijective correspondence with the set of all extended responses $\bar{f}: \Omega \to \Gamma$. Moreover, Σ realizes f iff $\bar{f} = \mathrm{Hog}$. Canonical realizations correspond dually to onto one-to-one factorizations of $A(\bar{f})$, which are unique in $\underline{\underline{V}}$. This results in an isomorphism $T: X_1 \to X_2$, which is easily shown to be a system isomorphism. (The same argument proves that in fact canonical realizations are initial for observable realizations and final for reachable ones.) □

In the case of interest, $\underline{\underline{V}}$ = k-algebras, a finiteness theory complements the above results. Specifically, for a polynomial response map f and a finite input sequence w we introduce the observable f^w, which maps each input sequence w' into $f(w'w)$ = output corresponding to first input w', then w.

When $Y \subseteq k^p$, each f^w induces p functions $U^* \to k$. The observation algebra $\underline{\underline{A}}_f$ is then the k-algebra (with pointwise operations) generated by all the f^w, w in U^*. This is an integral domain when U is irreducible, e.g. if $U = k^m$; in this case we denote $\underline{\underline{Q}}_f$ = quotient field of $\underline{\underline{A}}_f$ = observation field of f. A system Σ with X_Σ irreducible will be finite-dimensional [resp. polynomial] when $\mathrm{trdeg}_k(A(X_\Sigma))$ is finite [resp. $A(X_\Sigma)$ is a finitely generated k-algebra]. By construction, for U irreducible:

(5.3) THEOREM: Σ_f is finite-dimensional [resp. polynomial] iff $\mathrm{trdeg}_k(\underline{\underline{Q}}_f)$ is finite [resp. $\underline{\underline{A}}_f$ is finitely generated].

It is easy to generalize the results in SONTAG [1976 a] to prove that finite realizability is equivalent to the existence of a high-order input/output difference equation, and to give a Hankel-matrix-like rank condition for finite realizability. As there, various sufficient conditions (e.g. f satisfying a recursion) ensure finite generability of $\underline{\underline{A}}_f$. In the general case, Σ_f is only defined obstractly, the finite-dimensional case corresponding to locally rational transitions. Abstract (rather than algebraic) observability, or even the weaker notion of each generic pair of states being distinguishable (quasi-observability) forces only birational equivalence of canonical systems; this suggests the (so far open) question of finding quasi-reachable and quasi-observable but polynomial realizations.

(6) BOUNDED CASE:

Bounded input/output (and response) maps f are those for which there is an integer $d = d(f)$ bounding the degrees to which $u(j)$ can be raised in calculating $y(t)$, for all j and $t > j$. A natural class of realizations here is that of

(6.1) DEFINITION: A state-affine polynomial system Σ has $X = k^n$, and $P(x,u)$, $h(x)$ affine functions (linear + translation) of x. Σ is (affine) span-reachable if and only if the smallest linear manifold containing the reachable set of Σ is k^n, Σ is span-canonical iff both span-reachable and observable.

(6.2) THEOREM: Every bounded polynomial input/output map has a span-canonical state-affine realization. If Σ_1, Σ_2 are two such realizations, there is a system isomorphism $T: \Sigma_1 \to \Sigma_2$ with T an affine map.

PROOF: The "Behavior matrix" in SONTAG [1976 b] now has both rows and columns indexed by J^*, with the notations developed there. (In the equilibrium case, not all columns are needed!) The shift realization is still valid; now the state-space is the affine span of the columns of $\underline{B}(f)$. This gives a constructive algorithm to find realizations. For the uniqueness part, it is enough to prove that there is, for any Σ_1 span-reachable and Σ_2 observable realizations, a unique affine morphism $T: \Sigma_1 \to \Sigma_2$. Indeed, if $x = \sum a_i \hat{P}_1(x_0^{(1)}, w_i)$ is in X_1, $\sum a_i = 1$, ($\hat{P}$ = the extension of P to sequences), then we define $T(x) := \sum a_i \hat{P}_2(x_0^{(2)}, w_i)$. This map is well-defined by observability of Σ_2, and its domain is X_1 by span-reachability of Σ_1.

(7) FINAL REMARKS

We have treated various aspects of Volterra series realization theory. Similar methods have been applied in solving questions of control and observation for polynomial systems (SONTAG and ROUCHALEAU [1975], SONTAG [1978]). Generalizations to the case of nonaffine varieties as state-spaces are being currently studied. The main theoretical and practical difficulties with the present theory are to a great extent related to the behavior of the category of schemes with respect to one-to-one and/or onto morphisms. This has recently led to a re-evaluation of the Volterra series approach to nonlinear systems, and to the definition of totally new contexts for realization and control. One such context is that of systems defined over categories of piece-wise linear (PL) or continuous-PL (splines of order one) maps. These categories are much better behaved in regard to the standard categorical realization methods. Both classes, PL and CPL - systems, seem very natural both in modelling biological systems, where threshold effects are abundant, and from an implementation viewpoint, since microprocessor technology gives an easy way to design PL controllers. Research in this area is currently in progress.

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ABSTRACT REGULATION OF NONLINEAR SYSTEMS: STABILIZATION, Part II *

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ABSTRACT

This paper continues the study of the following problem: given a system S, find necessary and sufficient conditions for the existance of a regulator which stabilizes S (using only available state measurements). When S is defined by analytic differential equations, these conditions assume a particularly simple form. For bilinear (and hence also for linear) systems, they reduce to stabilizability and detectability. An example shows that, for more general systems, stabilizability and detectability are not sufficient.

1. INTRODUCTION

This paper constitutes the second part of [7]. The reader is referred to [7] for details of the relevant notations, definitions, and statements of the main theorems. Due to the length of Part I, the proofs of a number of results could not be included there. The objective of this more "technical" part is to provide details of those proofs omitted from [7], as well as to discuss an example which illustrates why the conditions in [7] cannot be weakened in the general case. To set the appropiate context, however, we shall first recall the following notions which were introduced in [7]; a short nontechnical description is given for each of them:

<u>Stability</u> ([7]2.9-2.10): the i/o map f is stable provided that finite support inputs produce asymptotically stable outputs.

<u>Regulability</u> ([7]2.16): the plant P with initial state "0" is regulable provided that there exists another initialized system (Q,0) such that the interconnection P*Q is well-posed and the closed-loop map from disturbances to output objectives is stable.

<u>0-detectability</u> ([7]2.23): the (positively invariant) subsystem [0/0] is asymptotically stable, where [0/0] consists of those states which are indistinguishable from x=0 under the zero input.

<u>Indy-asycontrollability</u> ([7]2.24): states can be driven asymptotically to the origin, with controls that depend only on ind(istinguishabilit)y classes. [<u>Asycontrollability</u> ([7]2.25) when controls are allowed to depend on the state itself.]

<u>Preregulability</u> ([7]2.26): 0-det. + indy-a.c.

<u>Indy-detectability</u> ([7]2.27): indy classes can be aproximately identified, using input output data, to any desired degree of accuracy.

The main results obtained in part I were: ([7]2.29) regulability = indy-detectability + preregulability; ([7]2.30) for analytic plants, indy-detectability always holds; and ([7]2.31) for state-affine --hence in particular for bilinear [1,2,3] plants-- preregulability = 0-detectability + asycontrollability (in other words, indy-a.c. is in that case a consequence of the latter two properties). For a study of piecewise linear implementations, see the related paper [6]; the sufficient conditions needed there, however, are somewhat stronger than the above. The notations in the this paper are as in Part I.

2. PROOFS.

The necessary part of ([7], Theorem 2.29) required the following technical lemma for establishing ([7], Propositions 3.2-3.4); its proof was not given in [7].

(2.1) LEMMA. Consider P*Q, and assume that v=0. Then, for each k,e > 0, and for each x' in X', there exist positive numbers d(e), m(k,x'), and T(k,e,x') such that: (a) if #(x)<d(e) and x'=0, then the closed loop trajectories have #(x(.))<e and #(u)<e, and (b) if #(x)<k and x' is reachable, then (i) #(x(t))<e and #(u(t))<e for all t $\geq$ T=T(k,e,x'), and (ii) x(.), y, u are in B(m(k,x')).

PROOF. Let e>0 be given. By [7]2.10, there are e'>0 and T>0 such that #(out[(0,0)/v]) < e whenever v is in B(e';T). Since P is metric, and using the local compactness of X and <u>V</u>, the map sending the triple ($\not d$,u,x) into the trajectory ϕ(.;x,(u,$\not d$)) is continuous, assuming the compact open topology is used for state trajectories. Thus, ϕ(t;0,0,0) $\doteq$ 0 implies that there exists for the above T an e' small enough so that #(x(t))<e for t$\leq$T if v is as above and u=0. We assume that e' satisfies also this property. By [7]2.22(i), for this e' there is a d such that, whenever x(.) is a trajectory which is in B(d) for t<T, there is some $\not b$ in B(e'/2;T) such that h(x(t),$\not b$(t)) = 0 for t<T. By [7]2.21(i), this d can be chosen small enough that for each x in B(d) there is an $\not d$ in B(e'/2;T) such that ϕ[T,0] sends $\not d$ to x. Further, d can be taken less than e. We let d(e):= this last d. Assume now that x is in B(d(e)), and pick $\not d$ as above. The continuity argument on ϕ gives that #(x(t))<e for all t$\leq$T. Thus there is a $\not b$ as above such that $\bar{y}$(t) = h(x(t),$\not b$(t)) = 0 for t<T. Let v:= ($\not d$,$\not b$). Then, if x(0)=0, x'(0)=0, all future x(t) (=w(t)) and u(t) are in B(e), and y|T|0 = 0, x(T) = x. It follows that x'(T)=0. Thus ϕ*(t;(x,0),0) = ϕ*(t;ϕ*(T;(0,0),v),0) = ϕ*(t+T,(0,0),v|T|0), and it follows that (a) holds.

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We now prove (b). We first establish that for any k and any reachable x' there exists a $c = c(k,x')$ and a T such that, whenever $\#(x)<k$, there is a v in $B(c)$ such that $\phi^*(T;(0,0),v) = (x,x')$. Since x' is reachable in Q, there is a y in its input signal space and a T such that $\phi'(T;0,y) = x'$. Let $u(t) := h'(x'(t),y(t))$ for $t \le T$. By [7]2.21(ii), there is for this k a $c>0$ such that, for each x in $B(k)$ there is an α in $B(c/2)$ so that $\phi(T;0,u,\alpha) = x$. By [7]2.4 applied to P/0, for each such a c there is a k' such that all intermediate $x(t)$ are in $B(k')$, when the input to P has α in $B(c/2)$ and u fixed as above. By [7]2.22(ii), then, c can be chosen large enough that, for any trajectory $x(.)$ starting in $B(k')$, there is a β in $B(c/2)$ with $h(x(t),\beta(t)) = y(t)$ for $t<T$. This $c = c(k,x')$ satisfies the desired property.

Assume now that e,k,x' are given. Choose a $T = T(x')$ as $T + T'$ satisfies the required properties. Let $N = N(c,T,T')$ be such that all of $x(t)$, $u(t)$, $y(t)$, are in $B(N)$ whenever $t \le T+T'$, v is in $B(c;T)$, and initial state = 0 (definition of "system" applied to P*Q). To obtain m, let T' be obtained for (say) $e=1$, and let $m(k,x')$ be larger than $N(c,T,T')$ and 1. This completes the proof of 2.1. */

We turn now to the sufficiency part of Theorem [7]2.29. The following lemmas were left without proof in Part I.

(2.2) LEMMA. Given a compact subset Z of X, and an open subset N, there exists an open set N' such that (i) N' is saturated rel Z, and (ii) for any indy class L such that N contains $L \cap Z$, N' satisfies $L \cap Z \subseteq N' \cap Z \subseteq N \cap Z$.

PROOF. Consider the restriction o to Z of the projection [.]. Since Z is compact and X/I is Hausdorff, o is a <u>closed</u> (continuous) map. Let B be the complement of N!!Z in Z. This set is closed in the relative topology of Z. Thus [B//Z] (=preimage under o of $o(B)$) is also closed in Z, and hence in X. Then $N' :=$ complement (in X) of [B//Z] satisfies the required properties. */

(2.3) LEMMA. Each of the functions $d(.)$ appearing in definitions [7]2.23, [7]2.24, and [7]2.27 may be assumed to be continuous and strictly increasing.

PROOF. Note that any function d' with $d'(e)<d(e)$ for all $e>0$ also satisfies the conditions in each case. Note also that, when $e'>e$, d can be redefined if desired at e' by $d'(e') := d(e)$, again without affecting the validity of the conditions. Given d, then, define first d' by: $d'(e) := d(1/n)$ in the interval $[1/n, 1/(n-1))$, for each $n>1$, and $d'(e) := d(1)$ otherwise. Now replace d' by any continuous strictly increasing function majorized by the piecewise constant d'. */

Only part of the following lemma will be needed later, but the full statement is of interest in itself. Note the difference with [7]2.24: T is now independent of L, and $x(t)$, $u(t)$ are in $B(r)$ <u>for all</u> $t \ge T$; on the other hand, u depends now on k and r. Only indy-asycontrollability is needed in 2.4-2.5.

(2.4) LEMMA. For each $k',r',e > 0$ with $r'<k'$, $e<k'$ there are positive $d'(e)$, $c(k')$, and $T(k',r')>1$ such that, for each indy class L', there is an input u' with: (i) if x is in $L' \cap B^*(d'(e))$ then both $\phi(t;x,u')$ and $u'(t)$ are in $B(e)$ for all t, and (ii) if x is in $L' \cap B^*(k')$ then $x(t) = \phi(t;x,u')$ is in $B(r')$ for all $t \ge T'(k',r')$, and $\#(x(t))<c(k')$, $\#(u)<c(k')$.

PROOF. Define a function $j(e) := d(d(e)/2)$, where d is as in [7]2.24, assumed continuous and strictly increasing via 2.3. Let also $d'(e) := j(e)/2$. For each class $L \ne [0]$, let $e(L)$ be the inf of the e such that $L \cap B^*(j(e))$ is nonempty. Since L is closed, $e(L) \ne 0$. Pick k',r',e, and a class L.

Assume first that $L \ne [0]$. Apply [7]2.24 to L, to obtain an input u as there. Let $r := d(r')$, $k := k'+1$, and pick T as in [7]2.24(ii). Pick a number c larger than k', larger than the (finite) number $\#(u|T|0)$, and larger than all possible values $\#(\phi(t;x,u'))$ for all $t \le T$, x in $B^*(k')$ and $\#(u') \le \max\{\#(u|T|0), k'\}$. Since $\phi[T,u]$ is continuous, there is a d'' such that any x,z at distance $< d''$ get mapped to x',z' at distance $< d(e(L))/2$, for x,z in $B^*(k)$. Let m be a number less than both d'' and $j(e(L))/2$. Note that $\phi[T,u]$ maps $E := L \cap B^*(k')$ into $B(d(r'))$. Let N be an open set which contains the former and which is also mapped into $B(d(r'))$. We may assume that N is contained in the m-ngbd of E. By 2.2, we may further assume that N is saturated rel $B^*(k')$. Consider any indy class L' intersecting N at a nonempty set F. Let G be the image of F under $\phi[T,u]$. Since $[x] = [z]$ always implies that $[\phi(t;x,u)] = [\phi(t;z,u)]$, there is a (unique) indy class H containing F. Let w be the input associated to H via [7]2.24. In particular, for any x in G, $\phi(t;x,w)$ remains in $B(r')$ for all t. Further, $\#(w) < r' < c$. Suppose that L' intersects some $B^*(d'(e))$, for some e, say at x. Since N is saturated rel $B^*(k)$, x is in N (because $d'(e)<e<k$). Thus there is some z in L with $\delta(x,z) < m < j(e(L))/2$. Since $\#(x) < d'(e) = j(e)/2$, it follows that $j(e(L)) \le \#(z) < j(e(L))/2 + j(e)/2$. Thus $e(L)<e$. It follows that L intersects $B(j(e)) = B(d(d(e)/2))$, and z is in the intersection. Also, this implies that u is in $B(j(e))$, so also in $B(e)$. By [7]2.24, $\phi(T;z,u)$ is in $B(d(e)/2)$. Since $\delta(x,z) < m < d-$, also $x' := \phi(T;x,u)$ is in $B(d(e))$. Thus $\#(\phi(.;x',w))<e$ and $\#(w)<e$. Concatenating u and w, one has for each L' intersecting N that the conclusions hold, except of course that T and c depend on the chosen L ($\ne [0]$).

We claim that the same local result holds for $L = [0]$. Again apply [7]2.24, to obtain T, with $r = d(r')$, $k = k'+1$. Note that now $u=0$. Since $E := [0] \cap B^*(k')$ is mapped by $u=0$ into $B(d(r'))$, there is by continuity on u and x an $a>0$ such that $\phi(T;x,w)$ is in $B(d(r'))$ whenever $\#(w)<a$ and x is in an a-ngbd of E. Let N be an open ngbd of E which is saturated rel $B^*(k)$ and which is contained in the $B(d(a))$-ngbd of E. Consider any indy class L' intersecting N at a nonempty set F. Choose for L' an input u as in [7]2.24. If L' intersects $B^*(d'(e))$, note that $d'(e) < d(e)$, so L' also intersects $B(d(e))$, and hence by [7]2.24 all future $x(t)$ and $u(t)$ are in $B(e)$. If x is in F, it is also in $B(d(a))$, so also $\#(u)<a$, and hence also $\phi(T;x,u)$ is in $B(d(r'))$, as wanted. With any c larger that a and larger than the $x(t)$ which may appear when starting in the closure of N, one has the conclusion for ngbds of $L = [0]$.

Cover the set $B^*(k')$ by the open sets $N(L)$ constructed for each L, and take a finite subcover. It is enough now to take c (resp., T) as the largest of the $c(L)$ (resp., all $T(L)$, and 1) associated to this subcover. */

(2.5) LEMMA. Let $T(.,.)$ and $d(.)$ be the functions in the statement of 2.4. For each $k,r,e > 0$ with $r<k$ and $e\leq 1\leq k$ there exists a $c(k)$, and for each indy class L intersecting $B^*(k)$ there exist an input u and an open set N, such that (i) $L\cap B^*(k) \subseteq N$, (ii) N is saturated rel $B^*(k)$, (iii) for any x in $N\cap B^*(k)$, $\#(\phi(T;x,u))<r$ and $\#(x(t))<c(k)$ for all $t\leq T$, (iv) if $e\leq e'\leq 1$ and x is in $N\cap B^*(d(e'))$ then $\#(x(t))<e'$ for $t\leq T$, and (v) $\#(u)<c(k)$.

PROOF. Let k,r,e,L be given. Pick u as in 2.4. Let M be the set of all x in X such that for each e' in $[e,1]$ the following property holds: $\#(x)\leq d(e')$ implies $\#(\phi(t;x,u))<e'$ for all $t\leq T$. An easy compactness argument (plus continuity of $d(.)$) gives that M is open. By 2.4, L is contained in M. Since $L\cap B^*(k)$ is mapped by $\phi[T,u]$ into $B(r)$, it has an open ngbd M' which is also sent to $B(r)$. Let N' be the intersection of M and M' and pick N satisfying (i),(ii), and with $N\cap B^*(k)$ included in $N'\cap B^*(k)$. Assume that x is in the first of these sets. Then x is in M', so it is mapped into $B(r)$. Define $c(k)$ as larger than $c'(k)$:= the "$c(k)$" of 2.4, and so that it bounds all values $\#(\phi(t;x,v))$ for $\#(v)<c'(k)$, $t\leq T$, and $\#(x)\leq k$. Thus (iii) holds. Since (v) holds by construction, it only remains to prove (iv). But this is immediate from the definition of M. */

(2.6) LEMMA. For each $k',k'',r,e,e' > 0$, $e\leq 1$, there are $T=T'(k',r,e)>1$, positive $c'=c'(k')$, $m'=m'(k',e,k'')$, $g=g(e')$, $b'=b'(k',e)$, and systems $Q=Q(k',r,e)$ with states $x'=x'(k',r,e)$ such that the following properties hold: (i) the interconnection $P*Q$ is well posed, and for $v=0$, $x(0)=x$, and $x'(0)=x'$ the following hold for the ensuing closed loop trajectories: (ii) if $\#(x)<g(e')$ for an e' in the interval $[e,1]$, then $\#(x(t))<e'$ and $\#(u(t))<e'$ for $t\leq T$, (iii) if $\#(x)<k'$ then $\#(x(T))<r$, and $\#(x(t))<c'$ and $\#(u(t))<b'$ for $t\leq T$, and (iv) if $\#(x)<k''$ then $\#(u(t))<m'$ for $t\leq T$.

PROOF. Let d be as in 2.5, and let d' be the function called d in [7]2.27. Define $g(e):= d'(d(e))$. Consider the detectors in [7]2.27, and let R be the $Q(k',d(e))$ there, $q = q(k',d(e))$. Let $X(R)$ be the state space of R. Using b from [7]2.27, let k be any number larger than $b(k',e)$, r, and 1. Let $c'(k'):= c(k)$, and let $b'(k',e)$ be any number larger than $b(k',e)$ and $c(k)$. Let $m'(k',e,k'')$ be any number greater than $c(k)$ and $m(k',e,k'')$. We shall define $T'(k',r,e)$ to be the sum of the $T(k,r)$ in 2.5 and of $T=T(k',d(e),o')$ (right-hand term is the estimate in [7]2.27), where the number o' will be constructed below.

We claim that there exists an open covering $\{N_i,\ i=1,\dots,n\}$ of $B^*(k)$ with all N_i saturated rel $B^*(k)$, inputs $\{u_i\}$ as in 2.5, and a map $\mathcal{I}$: $X' \to \{1,\dots,n\}$ such that: (1) $N_i\cap B^*(k)$ is mapped into $B(r)$ by u_i at time $T(k,r)$, with all the properties in 2.5 holding, and (2) when R is started at q and $x(0)$ is in $B(k')$, then $x(T)$ is in N_i for $i=\mathcal{I}(x'(T))$. We now prove this claim. Pick any indy class L, and apply 2.5 to get a corresponding N, u. Let $o(L)$ be such that the $o(L)$-ngbd of $L\cap B^*(k)$ is contained in N. Pick $o'(L):= o(L)/2$. Consider $B(L,o'(L))$ and apply 2.2 to this set, with $Z = B^*(k)$. Let $N'(L)$ be the saturated open thus obtained. Cover $B^*(k)$ by these $N'(L)$, and take a finite subcover $\{N_i'=N_i'(L_i),\ i=1,\dots,n\}$. By N_i, u_i we denote the N, u associated to the corresponding L_i. Let o' be a number which is less than 1 and than all the o_i'. This is the o' used above to obtain the time T that the detector R will be operated. Given any indy class L intersecting $B^*(k)$, one of the N' contains this intersection. Let $\mathcal{J}$: $X/I \to \{1,\dots,n\}$ be a choice map giving an index i for which this happens. Let $\mathcal{I}(z) := \mathcal{J}(\mathcal{I}(z))$. If R is now operated, with $x(0) = x$ in $B(k')$, one has $x(T)$ in $B(\mathcal{I}(x'(T)),o')$ by definition of indy-detectability. Thus $d(x(T),z)<o'$ for some z in $\mathcal{I}(x'(T))$. But $x(t)$ is always in $B(b(k',e))$, so $\#(z)\leq k$. By definition of $\mathcal{J}$, $\mathcal{I}(x'(T))\cap B^*(k)$ is contained in $B(L_i,o_i)$, where $i=\mathcal{I}(x'(T))$. Thus $d(x(T),L) < o_i$, and it follows that $x(T)$ is in N_i, as claimed.

We now define $Q = Q(k',r,e)$. The state space X' will be the set $(X(R)\cup\{1,\dots,n\})\times\mathbb{R}$ minus the set of all (p,s) with p in $X(R)$ and $s\geq T$. The state x' will be $(q,0)$. States of the type (p,t) in $X(R)\times\mathbb{R}$ will be said to be 'of type 1', the (i,t) in the rest of type 2. We let ϕ'', h'' denote the system functions of R. Then, for states of types 1 and 2 respectively:

$$
\begin{aligned}
(2.7)\quad \phi'(t;(p,s),y) &:= (\phi''(t;p,y),t+s) \quad \text{if } t+s<T\\
&:= (\mathcal{I}(\phi''(T-s;p,y)),t+s) \quad \text{else},\\
\phi'(t;(i,s),y) &:= (i,t+s),
\end{aligned}
$$

$$(2.8)\quad h'((p,t),\underline{y}):=h''(p,\underline{y});\ h'((i,t),\underline{y}):=u_i(t-T).$$

Then 2.7-2.8 define indeed a system, and the interconnection $P*Q$ is well posed: one needs only check all possible combinations of types of states and times $<T$ or $>T$. For instance, the signals u,y needed for the closed loop operation are constructed separately for $t<T$ (well posedness of $P*R$) and for $t\geq T$ (basically open-loop control in that interval). All the desired properties hold by construction. */

(2.9) LEMMA. Consider the system $S = P/0$. For each $k,e > 0$ there exist positive $\mathcal{J}(e)$, $\mathcal{M}(k)$ such that, for any t, x for which $out[x/0]|t|0 = 0$, the following properties hold: (a) if $\#(x)<\mathcal{J}(e)$ for some e, then $\#(\phi(t';x,0))<e$ for all $t'<t+1$, and (b) if $\#(x)<k$ for some k, then $\#(\phi(t';x,0))<\mathcal{M}(k)$ for all $t'<t+1$.

PROOF. By continuity of $\phi(t;.,0)$ there is a $g(.)$ such that $B(g(e))$ is mapped into $B(e)$ for all $t\leq 1$. By 0-detectability, there is a $d'(.)$ such that $\#(\phi(.;x,0)) < e$ whenever $[x/0] = [0/0]$ and $\#(x)<d'(e)$. Finally, let d be as in [7]2.27. Define $\mathcal{J}(.):= \min\{d'(.),\ d(g(.))\}$. Pick an x in $B(\mathcal{J}(e))$ and t arbitrary. Assume that $y|t|0 = 0$. By indy-detectability, either (i) $x(t')$ is in $B(g(e))$ for $t'\leq t$ or (ii) $x(t)$ is in $[0/0]$. In case (i), the choice of g gives the desired conclusion. If $[x(t)/0] = [0/0]$ then, since also $y(t')=0$ for $t'<t$, it follows that $out[x/0] = 0$, so x is itself in $[0/0]$. By 0-detectability, (a) holds. To prove (b), note that the estimate $\mathcal{M}(k)$ exists for x in $[0/0]$, and if x is not in this set, the estimate also exists by [7]2.27 plus the boundedness of $\#(\phi(t;x;0))$ for x bounded and $t\leq 1$. */

The main result for analytic systems ([7]2.28) follows from the next lemma, because the hypothesis hold as a consequence of the results in [8] or (for polynomial systems) [5] or (for bilinear) [4].

(2.10) LEMMA. Let P be a plant, and assume that the following properties hold for $S=P/0$, for some $T>0$: (i) for each e there is a u with $\#(u)<e$ and such that u determines final states, i.e., for any x,z in X, if $out[x/u]|T|0 = out[z/u]|T|0$ then $[\phi(T;x,u)] = [\phi(T;z,u)]$, and (ii) for any x in X, either $out[x/0]|T|0 \neq 0$ or $[\phi(T;x,0)] = [0/0]$. Then P is indy-detectable.

PROOF. Given $e>0$, pick $d(e)$ such that $\phi(t;x,u)$ is in $B(e)$ whenever $t\leq T$, $\#(x)<d(e)$, and $\#(u)<d(e)$. Such a d always exists by continuity, because 0 is an equilibrium state. Let $T(k,e,r):=$ constantly T, and let $m=m(k,e,k')$ be any number bounding $\#(\phi(t;x,u))$ for $t\leq T$, $\#(x)\leq k'$, and $\#(u)\leq 1$. Let $b(k,e):= m(k,e,k)$. The system $Q(k,e)$ will depend only on e; in fact, both the state space X and the transition map ϕ are even independent of e. Pick X as the set of initial segments of inputs (y,t) with $y|t|0 = y$. Let $\phi(t';(y,t),y'):= (y|t|(y'|t'|0), t+t')$. For each e, take an u as in the statement, such that $\#(u) < \min\{1,d(e)\}$. Define for $Q(k,e)$ the (strictly causal) output map $h(y,t):= u(t)$. Finally, choose $q(k,e):= (0,0)$, and let $\ell(y,t):=$ the indy class $[\phi(T;x,u)]$ characterized by $y|T|0$, if any such class exists, and arbitrary otherwise. (In the case to be considered, analytic systems, one may replace Y by a finite dimensional space.) The systems $Q(k,e)$ are all well defined, and the interconnection $P*Q(k,e)$ is well posed -trivially, since there is no feedback involved. The axioms (a), (b), (c), for indy-detectability are satisfied by construction. We now prove (d). Let t' be arbitrary and let $\#(x)<d(e)$. Assume that $out[x/0]|t'|0 = 0$. Let first $t'\geq T$. Then $[\phi(T;x,0)/0] = [0/0]$, so also $[\phi(t';x,0)/0] = [0/0]$, as wanted. If instead $t'<T$, then $\#(\phi(t;x,0)) < e$ for all $t\leq t'$, by definition of $d(.)$. To prove (e) the argument is analogous: just note that $\#(\phi(t;x,0))$ is bounded when x and t are both bounded. */

For state-affine systems, one wants to prove that weak preregulability, i.e. 0-detectability + asycontrollability, is equivalent to preregulability; this gives the standard linear characterization ([9]). It is proved in [7] that a weakly preregulable state-affine system is strongly locally indy-a.c. in the sense of Part I, so the wanted result follows from the following lemma.

(2.11) LEMMA. A weakly preregulable and strongly locally indy-a.c. system S is necessarily preregulable.

PROOF. We must prove indy-asycontrollablity. First modify d such that for no $e>0$ is $\#(x)<d(e)$ for an x not in N. Let $a>0$ be such that $B(a)$ is included in N. Let $k>0$, and pick any x in $B*(k)$. By asycontrollability, there are a u and a T such that $\#(\phi(t;x,u)) < a$ for all $t\geq T$. Using an argument as in 2.4, one can conclude that there as a whole ngbd of x that satisfies the same property, for a fixed $T(x)$, and for inputs u bounded by a constant dependent only on x. By compactness, one can find a $b(k)$ and a $T(k)$ such that all states in $B*(k)$ can be controlled into $B(a)$ in time $T(k)$ using inputs in $B(d(k))$. Let $b(.)$ and $T(.)$ be assumed increasing functions. Let $c(k)$ be a bound on the values $\#(\phi(t;x,u))$, for $\#(u)<b(k)$, $\#(x)\leq k$, and $t\leq T(k)$. Consider $T(.,.)$ from the definition of strong local indy-asycontrollability. Define now a new $T'(k,r):= T(k) + T(c(k),r)$.

Let now $L\neq 0$ be an indy class, and assume that L does not intersect N (otherwise, what follows is true by hypothesis). Let k be smallest possible such that L intersects $B*(k)$. Let x be any state in this intersection. Find u so that $\phi(T(k);x,u)$ is in $B(a)$. Then all of L is mapped into an indy class L' having a representative in $B(a)$. Thus (saturation) L' is included in N. There is then an input u' driving L' to the origin, and $u|T(k)|u'$ drives L to the origin. Condition (i) of [7]2.24 is irrelevant here, because L does not intersect N. Assume however that L intersects some $B(k')$, say at a state z. Necessarily, $k'>k$. So $b(k') > b(k)$, and thus $\#(\phi(T(k);z,u)) < c(k')$. It follows that, for the chosen input for L', z is mapped into $B(r)$ in time at most $T(k',r)$. This completes the proof. */

3. AN EXAMPLE

The example in this section shows that weak preregulability does not imply regulability, even for analytic systems. This example will obviously not be strongly locally indy-a.c.; its linearization, however, will be (even) asymptotically stable. The state space is the plane, and the inputs are scalar: $U = \mathbb{R}$. The equations are:

$$(3.1)\quad \dot{x} = (4x)\arctan(xz-1)(u+1) + 2x(1-z^2)(u^2+1)$$
$$\dot{z} = -z(u+1),$$

with output $y = z$. Note that solutions are defined for all $t>0$ and all admissible controls, because $|\arctan(xz-1)|$ is bounded.

This system is 0-detectable. Indeed, $[0/0] = [0] =$ line $\{z=0\}$, and on this subset the dynamics are given by the linear equation

$$(3.2)\quad \dot{x} = (2 - \pi)x,$$

which is stable. actually, more can be said about the case of the constant control $u=0$:

(3.3) CLAIM. There exists a number $a>0$ such that, whenever the product $|xz|<a$, the trajectory $\phi(t;(x,z),0)$ converges to the origin.

PROOF. Assume that the control $u=0$ is applied and consider a fixed $z(0)$. The z-coordinate of the solution is then $z(t)=\exp(-t)z(0)$, so that

$$(3.4)\quad \dot{x}(t) = x(t).f(x(t),t), \text{ where}$$
$$(3.5)\quad f(x,t) = 4\arctan(x.z(0).\exp(-t) - 1) + 2$$
$$\leq 4\arctan(x.z(0) - 1) + 2.$$

Since $4\arctan(-1) < 2$, there are $a,b>0$ such that $f(x,t) < -b < 0$ whenever $|x.z(0)| < a$, for all t. Thus $xf(x,t)$ and x have opposite signs whenever $|x.z(0)| < a$. It follows that the solutions of (3.4) converge to zero whenever $|x(0).z(0)| < a$, and the claim follows.*/

The system (3.1) is also asycontrollable. In order to establish this, we consider three cases: (1) $|z(0)| > 1$, (2) $0 < |z(0)| < 1$, and (3) $z(0) = 0$. In the first case, the constant control $u = -1$ results in a constant $z(t) = z(0)$, and

$$(3.6)\quad x(t) = \exp[4-4z(0)^2]\, x(0)$$

converges to zero. Thus there is a $T > 0$ with $|x(T)z(T)| = |x(T)z(0)| < a$, and one uses $u(t) = 0$ for $t > T$ (c.f. 3.3). In case 2, apply the constant control $u = -2$; the z-coordinate becomes unstable, so there exists a $T > 0$ at which time one is back in case 1. Finally, case 3 was treated earlier (apply $u=0$). This proves that all solutions of (3.1) converge to zero under appropiate controls. The local part of the a.c.

definition follows from the fact that the linearization of (3.1) at $x = z = 0$ is asymptotically stable, so that (3.1) is in fact asystable itself.

We must now prove that the given system is not indy-asycontrollable. This will follow from:

(3.7) CLAIM. There exists a number $e>0$ with the following property. If $x(0)z(0) > 1$, $z(0) > 0$, and if u is a control such that the ensuing trajectory $(x(t),z(t))$ converges to zero, then there is some $T \geq 0$ such that $|z(T)| \geq e$.

We first prove that this claim contradicts indy-asycontrollability. Assume that there is a function d as in the definition [7]2.24. Consider an $(\underline{x},\underline{z})$ such that $\underline{x} > 0$ and $\#(\underline{x},\underline{z}) < e$, where e is given by 3.7. Let L be the indy class of this point, i.e. the set of all $(x,\underline{z})$, x real. Pick an $x(0)$ such that $x(0).\underline{z} > 1$. Let u be any control as in [7]2.24; the corresponding trajectory $(x(t),z(t))$ with $z(0) = 0$ must then converge to zero. By (3.7), $|z(T)| \geq e$. But the same z-trajectory results from the ("small") initial state $(\underline{x},\underline{z})$, and this contradicts [7]2.24(i).

PROOF OF (3.7). Pick any number $0 < e < 1$ such that

(3.8) $-7 + 24c^2 - 16c^4 < 0$

for all $|c| < 2e$. For these c, and for all real b,

(3.9) $2(1-c^2)b^2 - b + (1-2c^2) > 0$.

Consider now a point $(\underline{x},\underline{z})$ in the positive branch of $xz=1$. The positively oriented normal to $xz=1$ at $(\underline{x},\underline{z})$ has the direction of $(1,\underline{x}^2)$. Let $r(x,z,u)$ denote the vector field in the r.h.s. of (3.2). Then $\langle r(\underline{x},\underline{z},b), (1,\underline{x}^2)\rangle$ equals

(3.10) $(1/\underline{z})[2(1-\underline{z}^2)b^2 - b + (1-2\underline{z}^2)]$,

for any b. It follows that the inner product is positive if $|\underline{z}|<2e$.

Assume now that $x(0)$, $z(0)$, and u are as in the hypothesis of (3.7), and let $(x(t), z(t)$ be the corresponding trajectory. Thus $x(t').z(t') < 1$ for some $t' > 0$. Assume that $|z(t)| \leq e$ for all t. There is then a continuous control u for which $x(t').z(t') < 1$ and $|z(t)| \leq 2e$ for t between 0 and t'. Let T be the largest $t < t'$ such that $x(t).z(t) = 1$. Since $(x(t),z(t))$ is in $(xz<1)$ for $T < t < t'$, it follows that $r(x(T),z(T),u(T))$ exists and points in the direction of $(xz<1)$. But this contradicts the choice of e.*/

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